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1 Introduction

Vector autoregressions (VARs) are extensively employed in economics to model multivariate time-series data. For instance, when identifying economic shocks, the initial step involves specifying the VAR model by selecting the relevant variables and determining the appropriate number of lags (Hamilton, 1994). Once this model is defined, structural identification methods—such as Cholesky decomposition and sign restrictions—allow researchers to isolate individual shocks and trace their effects on each variable over time. Moreover, because VARs account for potential interactions and feedback loops among variables, they are also powerful tools for economic forecasting (Lütkepohl, 2005).

Traditionally, in economics, the choice of lag length in a VAR model is somewhat arbitrary, often guided by intuition, data frequency, or trial and error, with adjustments made based on how well modeling outcomes capture expected dynamics (see, for instance, Bachmann et al. (2022) and also Bauer and Swanson (2022)). While more formal approaches exist, such as the Akaike Information Criterion (AIC) and its variations, they rely heavily on judgment, are often not transparent or reproducible, and lack a coherent probabilistic interpretation, making it challenging to quantify the uncertainty associated with lag selection (Lambert (2018); Choi (2011)). Bayesian approaches, on the other hand, allow for incorporating prior beliefs explicitly and provide a probabilistic framework.

A possible approach within the Bayesian framework involves several systematic steps to determine the appropriate lag order in a VAR model. Initially, researchers select a level of overparameterization — i.e., the number of lag orders under consideration — ensuring that the model is sufficiently flexible to capture the data dynamics without becoming excessively complex. Following this, researchers specify a prior distribution incorporating a shrinkage profile that aligns with prior beliefs about the relative importance of different lag lengths. After Bayesian updating, lags that are effectively zero get discarded.

Typically, researchers assign higher prior probabilities to lower lag orders to reflect the expectation that fewer lags are sufficient to capture the underlying dynamics of the data (Hamilton, 1994). This belief aligns with prior distributions with an increasing shrinkage profile, which impose progressively higher penalties on more distant lags. Examples of such priors comprise the Cumulative Shrinkage Process Prior (CUSP) introduced in (Legramanti et al., 2020) and the Multiplicative Gamma Process prior introduced in Bhattacharya and Dunson (2011), which is the primary focus of this thesis.

While leveraging a prior distribution to determine the optimal lag order in a VAR model offers compelling advantages, this approach can present substantial challenges. One is that

implementing those techniques requires considerable computational resources for potentially marginal improvements over more traditional techniques (Koop, 2013). The other, more relevant, is specifying a prior distribution over the parameter space may be difficult in empirical settings (Binks et al., 2023).

Researchers in economics, as well as other fields, commonly assume that Gaussian data is stationary. Stationarity implies that the statistical properties of the series, such as mean and variance, remain constant over time (Franses & van Dijk, 2014). In technical terms, stationarity constrains the autoregressive coefficients to lie within a subset of the parameter space known as the stationary region, which prevents the time series from exhibiting explosive behavior. Because the geometry of this stationary region becomes very complex in multidimensional settings, prior specification over this region becomes difficult (Binks et al., 2023).

As a solution, Heaps (2023) propose an unconstrained reparametrization of the stationary model, which enables specifying a prior invariant concerning the order of the components in the observation vector. The reparametrization is a two-step procedure wherein the original model parameters get mapped to a set of partial autocorrelations matrices, and the individual components of those matrices get rescaled from $[0, 1)$ to the positive real line. Notably, under this reparametrization, implementation is routine, as Markov chain Monte Carlo (MCMC) methods for computational inference operate over an Euclidean space.

Building on this reparametrization, Binks et al. (2023) pioneer a Bayesian inference framework to determine the order of the stationary VARs using a Multiplicative Gamma Process prior and inferential procedure described previously. Vetting the framework, they simulate ten independent samples of size $N = 1,000$ for VAR processes of dimension $m \in \{1, 3, 5, 7\}$ and orders $p \in \{1, 2, 3, 5\}$, contrasting the actual lag order to that the framework proposes. For nearly all combinations of m and p and experiment, they find that the median of the posterior distribution of the order is the actual order.

Notwithstanding these promising results, simulations with varying higher lag orders suggest a significant degradation in performance, consistently resulting in an underestimation of the actual order of the process. Investigating the implicit prior over the effective lag order led to the conclusion that, most likely, the original specification of the Multiplicative Gamma Process leads to excessive shrinkage. This finding motivated the development of a modified prior that extends the original Multiplicative Gamma Process by imposing a prior Gamma distribution on the local shrinkage rates, which are constant under the original specification. Theoretically, this should allow for more tailored regularization than the fixed rate shrinkage approach since it enables the model to adjust shrinkage rates across local precision

parameters according to the data's specific characteristics, ultimately leading to improved performance in higher-dimensional settings. Results from simulation studies presented in this thesis further support this theoretical understanding, as the modified prior shows a marked improvement in estimating the actual lag order in settings with many lags.

In what follows, the literature review provides an overview of frequentist and Bayesian approaches to model selection, focusing on different shrinkage priors. The theoretical framework explains the Multiplicative Gamma Process and introduces its proposed extension, focusing on how different hyperparameter choices affect shrinkage outcomes. Next, the Methodology section outlines the Bayesian inference framework for selecting the lag order, which follows the approach of Binks et al. (2023), details the Monte Carlo algorithm used to sample from the prior distribution over the effective lag order, and outlines the simulation design and performance evaluation process used to compare the original and modified MGP specifications. The Simulation Results section then presents the outcomes of these evaluations. Finally, the thesis concludes with a discussion about findings, limitations, and avenues for future research.

2 Literature Review

A Vector Autoregression (VAR) is a linear model comprising n equations and n variables, where each equation explains a variable using its past values along with the past values of the other $n - 1$ variables (Franses & van Dijk, 2014). More formally, a $VAR_n(p)$ process of order p and n endogenous variables can be expressed as

$$\mathbf{y}_t = \mathbf{c} + \mathbf{M}_1 \mathbf{y}_{t-1} + \mathbf{M}_2 \mathbf{y}_{t-2} + \cdots + \mathbf{M}_p \mathbf{y}_{t-p} + \boldsymbol{\epsilon}_t, \quad (1)$$

where $\mathbf{y}_t$ is the vector of endogenous variables at time t , $\mathbf{c}$ is the vector of constant terms, each $\mathbf{M}_i$ is a coefficient matrix for lag i , and $\boldsymbol{\epsilon}_t$ represents the vector of error terms.

Because they effectively capture complex dynamics across multiple time series within a simple framework, vector autoregressions are widely used in economics for forecasting and structural inference (Stock & Watson, 2001). Within this context, and especially when data are Gaussian, a common assumption is that of stationarity (Franses & van Dijk, 2014). Although this assumption can be unrealistic due to systematic variations like seasonality and trends, stationary VARs are often foundational elements within more complex models, such as those applied to differenced data in integrated models or as components in state-space models thought to be mean-reverting (Binks et al., 2023). See, for example, Camarero et al. (2020), where a state-space model incorporating stationary VARs is applied to a 17-country panel dataset to investigate the Feldstein–Horioka puzzle.

Despite their widespread usage, estimating VAR models presents considerable challenges. The number of parameters increases quadratically with the number of time series components, leading to many parameters that must be estimated (Wang et al., 2022). Since models with many parameters require substantial data to ensure reliable parameter estimation, VAR models in settings with limited data — as is usually the case in macroeconomics — are vulnerable to unreliable estimates. Hence, for these reasons and others discussed later, being selective about the endogenous variables to include and order to consider is crucial. The next part of this review explores the main paradigms underlying model selection, emphasizing different techniques associated with each approach.

Modeling in economics targets a precise understanding of economic relationships and the development of accurate and robust projections. Within the overall modeling process, model selection plays a crucial role in achieving these goals, with the choice of predefined selection criteria naturally impacting the outcome of the selection procedure and, at the same time, implicitly embedding certain assumptions about the economic system under consider-

ation. Consequently, deliberately choosing the selection criteria is necessary to ensure that the implied economic assumptions accurately reflect beliefs or, at least, reasonably support modeling objectives. In what follows, this review analyzes the relationship between elemental principles underlying the formulation of model selection methodology and implied economic implications, offering a taxonomy of existing model selection methodologies.

One of the fundamental challenges in statistical modeling is achieving a balance between the goodness of fit of a model and its complexity (Franses & van Dijk, 2014). The idea is that when including additional parameters in a model, predicted values may better match the observed data, not because the model becomes better at capturing underlying patterns but because it is capturing noise. Specifically, the model may become overfitted to the observed data, leading to poor performance on unseen data and possibly deceiving conclusions about the relationship between variables.

While rooted in earlier scientific principles like Occam's Razor, Akaike (1998) first formalized the trade-off by introducing a systematic and quantitative method for implementing the balance through the Aikaike's Information Criterion (AIC). Let $\hat{\theta}$ be a vector of dimension k containing the maximum likelihood estimates of the model parameters. The AIC value of a model of some data is then

$$AIC = -2 \sum_{i=1}^N \log f(x_i | \hat{\theta}) + 2k, \quad (2)$$

with $x_i (i = 1, 2, \dots, N)$ the i —th independent realization of a random variable X following the distribution with density function $f(x|\theta)$.

The AIC posits that the most parsimonious model among the candidate models — i.e., the one better able to fit the data while remaining adequately simple — should have the lowest AIC value in the set. Indeed, the first term of Equation 2 rewards model fit via the likelihood function, and the second discourages complexity through introducing a penalty that is an increasing function of the number of parameters. Furthermore, since the AIC values themselves are not interpretable due to the unknown interval scale, they offer only a measure of relative performance for competing models.

One of the most significant drawbacks of the AIC is that it tends to favor overly complex models when there are too many parameters compared to the sample size. As explained in Burnham and Anderson (2002), the AIC estimates the relative, expected Kullback-Leibler distance, which measures the expected information loss of approximating the actual data-generating process with some candidate model relative to other candidate models. While

this estimator is asymptotically unbiased under certain conditions, the penalty term may not be sufficient to offset the improvement in fit that additional parameters provide in small samples, leading to an underestimation of the information loss. As a revision, (Sugiura, 1978) proposed the corrected Akaike's Information Criterion (AICc), which introduces an additional bias correction to the original formulation that converges to zero as the sample size increases:

$$\text{AIC}_c = \text{AIC} + \frac{2K(K+1)}{N-K-1}. \quad (3)$$

Criteria estimating Kullback-Leibler divergence operate on the principle that the functional form of dependence between the dependent variable and the regressor is infinitely complex, so the goal of model selection should be to identify the best low-dimensional approximation of the model. Conversely, dimension-consistent criteria assume that a model that completely captures reality exists, aiming instead at identifying this model. An example of such criteria is the Bayesian Information Criterion (BIC) introduced in Schwarz (1978), which selects the model M_j in the candidate set $M_k (k = 1, \dots, K)$ with the highest posterior probability. Assuming that all models are a priori equally likely, maximizing the posterior probability is equivalent to maximizing the likelihood of the data marginalized over the model's parameter space. Asymptotically, this is equivalent to choosing the model that minimizes the BIC value:

$$\text{BIC} = -2 \sum_{i=1}^N \log f(x_i | \hat{\theta}) + k \log(N). \quad (4)$$

While the theoretical justifications underlying the derivations of both criteria differ significantly, the primary functional difference lies in the penalty term — the penalization BIC employs is $k \log(N)$ while that of AIC is $2k$. Accordingly, for any $N > 7$, BIC penalizes complexity more severely than the AIC, leading to the selection of more parsimonious models, especially in large-sample settings. Besides, as Vrieze (2012) explains, asymptotic properties also differ. If the correct model is among the candidates, the probability that the BIC determines the correct model converges to one as the sample size increases relative to the number of estimable parameters — a property known as consistency. Meanwhile, the AIC is asymptotically efficient, meaning that it selects the predictively optimal model according to the mean squared prediction error.

The asymptotic properties of AIC and BIC underscore a broader discussion about the role of model selection in achieving modeling objectives. As discussed by Shmueli (2010) and numerous other sources, statistical modeling serves two primary purposes — inference and

prediction. In economics, inference typically involves using association-based models applied to observational data to test causal hypotheses about phenomena of interest. Meanwhile, the goal of predictive modeling in economics is to predict some output value based on input values. Because the AIC minimizes the mean squared prediction error, the consensus is that it is better when the goal of the modeling task is prediction (Anderson, 2008). In contrast, BIC's consistency implies that it is best when the modeling aims to describe a process by selecting the most meaningful predictors, a task more closely related to inference (Anderson, 2008).

Despite the inextricable link between prediction and inference, a model designed for prediction might not always offer meaningful insights into causal mechanisms. That is specifically true for some popular machine learning algorithms. Consider, for instance, traditional random forests. Because they rely on ensemble learning — i.e., aggregation of predictions of a group of predictors — and can capture complex non-linear patterns, random forests can predict with excellent accuracy (Zhang & Ma, 2012). Nevertheless, auxiliary outputs that random forests produce, such as measures of relative feature importance, can at most suggest a correlation between the features and the outcome (Breiman, 2001).

Model selection in modern machine learning emphasizes a standard rigorous approach to ensure robust model performance on unseen data. As Géron (2022) explains, the dataset splits into some percentage for training and another for testing, with the training set further divided into a smaller training set and a validation set. When appropriate, for the potential methods hyperparameter tuning techniques like Grid and Random Search identify the best hyperparameters. Finally, evaluating the performance of the tuned models on the validation set, the best model follows from optimizing a specific performance metric. The more common performance metrics for regression tasks include the Mean Absolute Error (MAE), given by the average of the absolute differences between predicted and actual values, and Mean Squared Error (MSE), expressed as the average of the squared differences between predicted and actual values:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |\hat{x}_i - x_i|, \quad (5)$$

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (\hat{x}_i - x_i)^2. \quad (6)$$

Under the model selection framework described above, shuffling the data before splitting the dataset is standard for cross-sectional data to avoid potential biases. Yet, randomization is inappropriate for time-series data because maintaining the sequential order of the data points is crucial for preserving the temporal dependencies inherent in the data. A simple solution is to split the data chronologically. However, if the data exhibits time-varying trends or seasonality, this may lead to overfitting to specific periods. The better approach uses cross-validation techniques, which systematically partition the dataset into multiple training and validation sets. Examples appropriate for time series include rolling window cross-validation, which uses a fixed-size training window that shifts forward in time for each iteration, and expanding window cross-validation, which incrementally expands to include more historical data with each step.

The information criteria described above and machine learning frameworks are flexible, allowing for comparing many models. Notwithstanding, traditional information criteria like the AIC require the likelihood to be computable and complexity quantifiable, which is not straightforward for non-parametric models (Jullum & Hjort, 2017). On the other hand, the machine learning framework optimizes for prediction alone, meaning that for inferential purposes, they are helpful insofar as predictive accuracy translates into inferential conclusions for the chosen model (Géron, 2022).

Beyond the techniques described above, less flexible approaches are available for nested models—i.e., those obtained from imposing appropriate parametric restrictions or as limits of suitable approximations (Heij et al., 2004). As Lambert (2018) and other sources highlight, a popular class known as shrinkage methods adjusts parameter estimates by introducing constraints or penalties that pull them toward specific values. These techniques can shrink coefficients toward zero, effectively selecting a subset of variables and simplifying the model, and are interpretable from a frequentist and Bayesian perspective.

In the frequentist approach, shrinkage follows from adding penalty terms to the objective function used during parameter estimation. Examples include Lasso (Least Absolute Shrinkage and Selection Operator) (Tibshirani, 1996) and Ridge regressions (Hoerl & Kennard, 1970), which incorporate penalties based on the magnitude of the coefficients. Specifically,

Ridge regression adds a penalty term proportional to the L_2 norm of the parameters and Lasso proportional to the L_1 norm. Importantly, the frequentist framework optimizes this penalized loss function without involving prior distributions or probabilistic interpretations of the parameters.

Converse to the frequentist approach, the Bayesian approach to shrinkage involves incorporating prior beliefs about the parameters directly into the model through prior probability distributions (Lambert, 2018). By selecting priors that favor specific values—often centered around zero—Bayesian methods naturally shrink parameter estimates toward these values when updating beliefs with observed data. This allows for greater flexibility, as the choice of prior distribution can reflect specific knowledge or assumptions about the parameters. In addition, the Bayesian approach — as with the Bayesian paradigm in general — provides a full probabilistic description of parameter uncertainty through the posterior distribution, which enables direct and comprehensive quantification coefficient uncertainty.

A natural choice for a shrinkage prior is the Gaussian (Normal) distribution centered at zero, which encourages coefficients to remain small through its variance parameter. Notably, this is analogous to ridge regression, for which one interpretation is that of performing Maximum A Posteriori (MAP) estimation with a Gaussian prior on the coefficients (Brown & Griffin, 2005). Similarly, the Laplace (Double Exponential) prior corresponds to the frequentist Lasso, promoting sparsity by encouraging some coefficients to be zero (Park & Casella, 2008).

Note that the Gaussian and Laplace priors described above are not hierarchical; they apply directly to the coefficients with fixed parameters controlling the degree of shrinkage. In contrast, hierarchical priors introduce an additional layer of parameters, known as hyperparameters, which control the amount of shrinkage and follow from the data (Lambert, 2018). Explained differently, this hierarchical structure allows for adaptive regularization through data-driven shrinkage, where the model can automatically adjust the level of shrinkage based on the data characteristics.

A notable example of a hierarchical shrinkage prior is the horseshoe prior distribution introduced in Carvalho et al. (2010), recognized for its robustness in handling sparse signals. It combines local and global shrinkage through a hierarchical structure, enabling adaptive regularization by introducing hyperparameters that respond to data sparsity. Specifically, each coefficient β_i has an associated local shrinkage parameter λ_i that operates at the individual level, while a global shrinkage parameter τ adjusts the overall shrinkage across all coefficients. This hierarchical setup allows for selective shrinkage, where small coefficients (likely noise) are shrunk toward zero while larger coefficients (significant signals) remain

relatively unaffected. Following is its formal specification:

$$\beta_i \mid \lambda_i, \tau \sim N(0, \lambda_i^2 \tau^2), \quad \lambda_i \sim C^+(0, 1), \quad \tau \sim C^+(0, 1), \quad (7)$$

where $C^+(0, 1)$ denotes the half-Cauchy distribution.

This structure results in a unique shrinkage profile: an infinitely tall spike at zero enforces high shrinkage for coefficients near zero, encouraging noise reduction, while the Cauchy-like tails preserve the bigger coefficients, preventing the overshrinking typical in other priors. When considering model selection, these characteristics make the horseshoe prior suitable for high-dimensional variable selection. Logically, this is especially true in sparse linear regression, where many predictors may be irrelevant or near zero.

Notwithstanding, there are scenarios where the horseshoe prior may not be suitable. One significant case is when increasing shrinkage is necessary. Consider, for example, in time series modeling, selecting the order of terms in autoregressive (AR) models. For this task, a common underlying assumption is that later terms (higher orders) should exert progressively less influence on the system relative to more recent terms (Franses & van Dijk, 2014). In such cases, a prior that imposes an increasing shrinkage profile—i.e., a structure in which the degree of shrinkage applied to each coefficient grows progressively along the sequence of terms—is more suitable (Binks et al., 2023).

One example of an increasing shrinkage prior is the cumulative shrinkage process (CUSP) introduced in Legramanti et al. (2020). Specifically, this is a spike-and-slab shrinkage prior for which the spike probability is stochastically increasing and constructed from the stick-breaking representation of a Dirichlet process prior (Frühwirth-Schnatter, 2023). Formally, the prior distribution is defined for a sequence of parameters $\{\theta_h : h \in \mathbb{N}^+\}$, where each parameter θ_h is assigned a spike-and-slab distribution. The spike probability for each θ_h increases progressively, imposing cumulative shrinkage along the sequence. That is,

$$(\theta_h \mid \pi_h) \sim P_h = (1 - \pi_h)P_0 + \pi_h \delta_{\theta_\infty}, \quad (8)$$

$$\pi_h = \sum_{l=1}^h \omega_l, \quad \omega_l = v_l \prod_{m=1}^{l-1} (1 - v_m), \quad (9)$$

where $v_h \{v_h : h \in \mathbb{N}^+\}$ are independent $\text{Beta}(1, \alpha)$ variables while P_0 is a diffuse continuous distribution.

The stick-breaking construction of the Dirichlet process encourages shrinkage by concentrating probability mass around θ_∞ as the model complexity grows. As π_h increasingly converges to 1 with higher dimensions, this process effectively pulls redundant terms towards θ_∞ , reducing their influence in the model. At the same time, P_0 is the prior on the relevant parameters, typically a continuous distribution allowing flexibility for larger values of θ_h . δ_{θ_∞} represents a Dirac delta spike at θ_∞ , typically near zero. The sequence π_h uses a stick-breaking process with ω_l representing the break weights. As h increases, π_h accumulates towards 1, thus increasing the probability that later parameters are drawn from the spike distribution.

While effective in imposing increasing shrinkage across a sequence of coefficients, the CUSP prior is sensitive to the choice of its hyperparameters (Frühwirth-Schnatter, 2023). This sensitivity means tuning is often required to achieve the desired shrinkage effect, making implementation complex. Considering this, an alternative is the Multiplicative Gamma Process (MGP) prior from Bhattacharya and Dunson (2011), which introduces increasing shrinkage through a multiplicative structure of gamma-distributed random variables. This structure inherently induces a sequential increase in shrinkage without requiring intensive hyperparameter tuning, making it more robust across different settings (Binks et al., 2023). For a detailed explanation of the MGP prior, refer to Section 3.1.

Finally, within economics, the Minnesota prior (Litterman, 1986) addresses the forecasting challenges inherent in vector autoregressive (VAR) models, particularly when analyzing macroeconomic data with limited sample sizes. For a $\text{VAR}_m(p)$ process of dimension m and order p , the number of parameters to be estimated is m^2p for the autoregressive coefficients, m for the intercept terms, and $\frac{m(m+1)}{2}$ for the covariance matrix of the error terms, resulting in a total of $m^2p + m + \frac{m(m+1)}{2}$ parameters. This parameterization highlights the rapid increase in complexity as the number of variables m and the lag order p grow, underscoring the importance of regularization techniques like the Minnesota prior to ensure reliable and interpretable estimates in macroeconomic applications.

The Minnesota prior introduces an informative structure that shrinks parameters toward a simpler model, generally leading to stable, accurate forecasts that outperform those constructed from conventional VAR models when data is scarce (Giannone et al., 2013; Litterman, 1986). Formally, the Minnesota prior applies a Gaussian distribution to the coefficients in a VAR model, represented by the system:

$$y_t = C + B_1 y_{t-1} + \dots + B_p y_{t-p} + \varepsilon_t, \quad t = 1, \dots, T \quad (10)$$

where y_t is a vector of endogenous variables, B_i are coefficient matrices, and ϵ_t are error terms. The system above can be rewritten using matrix notation as

$$Y = X\beta + \epsilon \quad (11)$$

$$\epsilon \sim \mathcal{N}(0, \Sigma \otimes I_{T-p}). \quad (12)$$

The prior centers these coefficients around a random walk with drift by assuming a Normal distribution (Giannone et al., 2013):

$$\beta \mid \Sigma \sim \mathcal{N}(b, \Sigma \otimes \Omega), \quad (13)$$

with b representing the prior mean. This formulation prioritizes each variable's first lag, assuming recent data points hold the most predictive value, and emphasizes own-variable lags over cross-variable lags. Specifically, the impact of distant lags decreases systematically, with prior variance reducing according to an inverse-square rule for each lag order j (Litterman, 1986):

$$\text{Var}(\beta_j) = \frac{\sigma^2}{j^2}, \quad (14)$$

where σ^2 represents the variance of the first lag coefficient.

A significant drawback of the Minnesota prior and similar shrinkage techniques for reducing the dimensionality of the autoregressive coefficient is that they typically overlook the structure within the error covariance (Huber et al., 2024). As a solution, Huber et al. (2024) propose a new approach that departs from the conventional shrinkage on the Cholesky factor and directly imposes a prior—the stochastic block model (SBM) prior—on the precision matrix Ω . This approach accounts for the conditional dependencies between shocks in a networked form.

The SBM prior clusters variables into groups based on their likelihood of interaction, promoting a sparse structure where within-group relations remain strong while relations across groups weaken or disappear. This structure allows for more accurate predictions — evidenced by better density forecast accuracy, particularly in predicting complex networked relationships among macroeconomic variables, compared to traditional priors in simulations and forecast studies of U.S. data.

Formally, the model introduces a binary indicator $\delta_{i,j}$ that determines whether a relationship exists between shocks ε_i and ε_j , expressed as:

$$\omega_{i,j} \sim \delta_{i,j}N(0, \tau_{i,j,1}^2) + (1 - \delta_{i,j})N(0, \tau_{i,j,0}^2), \quad (15)$$

where $\omega_{i,j}$ represents the element in the precision matrix and $\tau_{i,j,1}^2$ and $\tau_{i,j,0}^2$ control the variance in the “slab” (non-zero) and “spike” (zero) components. By clustering shocks that show strong within-group relations, the model identifies sparse, interpretable network structures in the data, enhancing computational efficiency and forecast accuracy.

3 Theoretical Framework

3.1 The Multiplicative Gamma Process Prior

Within the context of Bayesian factor models for high dimensional data, the Multiplicative Gamma Process (MGP), proposed by Bhattacharya and Dunson (2011), facilitates adaptive dimensionality reduction by introducing a flexible shrinkage mechanism. It achieves this by imposing a structured sequence of global-local shrinkage priors for the loadings matrix in infinite factor models, allowing for progressive penalization of additional latent factors while controlling for the local precision of individual loading elements. By promoting heterogeneous shrinkage across parameters, the MGP provides enhanced flexibility compared to the uniform behavior imposed by strict global shrinkage, which in real-world applications can frequently lead to the over-penalization of significant loadings.

The adaptive mechanism of the MGP operates through a hierarchical prior structure. Let A_s represent the matrix controlling the coefficients in a Vector Autoregressive (VAR) model. Borrowing the notation from Binks et al. (2023), under the MGP each element $a_{s,ij}$ is modeled as:

$$a_{s,ij} \mid \lambda_{s,ij}, \tau_s \sim \mathcal{N}\left(0, (\lambda_{s,ij} \tau_s)^{-1}\right), \quad (16)$$

where $i, j = 1, \dots, m$ and $s = 1, \dots, S$. In this formulation, $\lambda_{s,ij}$ represents the local precision parameter for each element $a_{s,ij}$, and follows the prior:

$$\lambda_{s,ij} \sim \text{Gam}\left(\frac{a}{2}, \frac{a}{2}\right), \quad (17)$$

independently for each $i, j = 1, \dots, m$ and $s = 1, \dots, S$. Meanwhile, the precision parameters τ_s control for global shrinkage and are constructed through a multiplicative process:

$$\tau_s = \prod_{k=1}^s \delta_k, \quad \delta_1 \sim \text{Gam}(a_1, 1), \quad \delta_k \sim \text{Gam}(a_2, 1) \quad \text{for } k \geq 2, \quad (18)$$

with δ_k generated independently.

At the global level, the increasing shrinkage behavior of the MGP depends on the choice of hyperparameters $a_1 > 0$ and $a_2 > 0$. Since the global precisions τ_s are cumulative products of gamma random variables, $a_2 > 1$ guarantees that their expectations are increasing in s , thus ensuring that the variance of the individual entries $a_{s,i,j}$ converge to zero as the number of

than c_2 ensures that most local-parameter draws fall below 1, with the values decreasing further as the gap between c_1 and c_2 widens. Since the variance of the $a_{s,i,j}$ elements is inversely related to the local parameters, this configuration should help offset some of the global shrinkage. More detailed guidelines on how the gap between c_1 and c_2 affects the mitigation of over-shrinkage are in Section 5.

3.3 Implications for Lag Order Determination

The goal of using the MGP to select the lag order in Stationary Vector autoregression (VARs) is to apply increasing shrinkage on the parameters corresponding to higher lags. The underlying assumption is that lower lags (closer in time) contribute more significantly to prediction, while higher lags contribute less. By progressively shrinking the parameters associated with these higher lags towards zero, the MGP helps to identify the point where additional lags no longer provide meaningful information.

While theoretically interesting, determining the lag order of stationary VARs can be challenging. For a process to be stationary, autoregressive parameters must lie in the stationary region, which is usually geometrically complex, thus making prior specification difficult. To overcome this, Heaps (2023) developed a method for reparametrizing the autoregressive process by mapping the original parameters to a set of partial autocorrelations and then scaling singular values of each of these partial autocorrelation matrices from $[0,1)$ to the positive real line. This unconstrained reparametrization allows for prior specification, making it possible to enforce stationarity while selecting the appropriate model order in a Bayesian framework.

Binks et al. (2023), relying on this unconstrained reparametrization, presents a methodology for determining the order of stationary VARs by imposing the MGP prior distribution on the set of transformed partial autocorrelation matrices. In this approach, the point at which the partial autocorrelations become zero — i.e. when they fall below a pre-determined truncation threshold — determines the lag order of the process. Although initial simulation results for the method show promise, the analysis focuses exclusively on small lag orders. Extending the simulations to processes of larger lag orders reveals that the original MGP specification imposes excessive shrinkage, causing an underestimation of the actual lag order. This finding led to the development of the extended MGP prior, designed to reduce shrinkage and enable accurate identification of the lag order in settings with many lags.

4 Methodology

4.1 Bayesian Inference for VAR Lag order Determination

4.1.1 Overview

The Bayesian inference framework for determining the lag order in stationary autoregressive processes follows directly from Binks et al. (2023). The approach imposes an MGP prior distribution on matrices of transformed partial autocorrelations, leveraging the reparametrization proposed by Heaps (2023) that enables this. This section outlines the most relevant aspects of the methodology — borrowing the original notation — with the proposed modifications to address issues in settings with large lag orders presented in the following sections.

4.1.2 Reparametrization of stationary VARs

Let the m -variate process $\{y_t\}$ be modeled as a zero-mean vector autoregression of order p , denoted $\text{VAR}_m(p)$:

$$y_t = \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + \varepsilon_t, \quad (21)$$

where the errors ε_t represent a sequence of uncorrelated, zero-mean multivariate normal random vectors, $\varepsilon_t \sim N_m(0, \Sigma)$. The model parameters include the autoregressive coefficient matrices $\phi_i \in M_{m \times m}(\mathbb{R})$ for $i = 1, \dots, p$, collectively denoted as $\Phi \in M_{m \times m}(\mathbb{R})^p$, and the error variance matrix $\Sigma \in S_m^+$. In this context, $M_{m \times n}(V)$ refers to the space of $m \times n$ matrices with entries in V , and S_m^+ denotes the space of $m \times m$ symmetric, positive definite matrices.

Using the backshift operator B , where $B^k y_t = y_{t-k}$, Equation 21 can be expressed as

$$\varepsilon_t = (I_m - \phi_1 B - \cdots - \phi_p B^p) y_t = \phi(B) y_t, \quad (22)$$

with I_m as the $m \times m$ identity matrix, and $\phi(u) = (I_m - \phi_1 u - \cdots - \phi_p u^p)$, the characteristic polynomial. Stationarity holds if and only if the roots of the characteristic polynomial $\det(\phi(u)) = 0$ lie outside the unit circle in the complex plane. This condition defines the stationary region $C_{p,m}$, the subset of $M_{m \times m}(\mathbb{R})^p$ where stationarity conditions are satisfied.

As p or m increases, the geometry of the stationary region $C_{p,m}$ becomes increasingly complex, making prior specification difficult. Besides, designing an efficient MCMC sampler targeting the distributions constrained to this region is challenging. The reparametrization

proposed by Heaps (2023) addresses these issues by mapping the constrained parameters onto an unconstrained space, achieved through a two-step bijective mapping process.

The first step maps the original model parameters $(\Sigma, \Phi) \in S_m^+ \times C_{p,m}$ to a new parameter set $\{\Sigma, (P_1, \dots, P_p)\} \in S_m^+ \times \mathcal{V}^p$, where $\mathcal{V}$ refers to the subset of matrices in $M_{m \times m}(\mathbb{R})$ with singular values less than one. Each matrix P_{s+1} represents the $(s+1)$ -th partial autocorrelation matrix, that is, the conditional cross-covariance between y_{t+1} and y_{t-s} , given $y_t, \dots, y_{t-s+1}$. To standardize P_{s+1} , one uses the following expression:

$$\Sigma_s^{-\frac{1}{2}} \text{Cov}(y_{t+1}, y_{t-s} \mid y_t, \dots, y_{t-s+1}) \Sigma_s^{*- \frac{1}{2}}, \quad (23)$$

where $s = 0, \dots, p-1$. Here, Σ_s and Σ_s^* represent the conditional variances defined as

$$\Sigma_s = \text{Var}(y_{t+1} \mid y_t, \dots, y_{t-s+1}) \quad \text{and} \quad \Sigma_s^* = \text{Var}(y_{t-s} \mid y_t, \dots, y_{t-s+1}), \quad (24)$$

with $\Sigma^{\frac{1}{2}}$ representing the symmetric matrix-square root.

The second mapping transforms each partial autocorrelation matrix $P \in \mathcal{V}$ to an unconstrained square matrix $A \in M_{m \times m}(\mathbb{R})$ as follows:

$$A = (I_m - PP^\top)^{-\frac{1}{2}} P. \quad (25)$$

The singular value decomposition of P takes the form $P = U \text{diag}(r_1, \dots, r_m) V^\top$, where $1 > r_1 \geq r_2 \geq \dots \geq r_m \geq 0$. Consequently, the corresponding factorization of A is,

$$A = U \text{diag}(\tilde{r}_1, \dots, \tilde{r}_m) V^\top, \quad (26)$$

where $\tilde{r}_i = \frac{r_i}{(1-r_i^2)^{1/2}} \geq 0$ for $i = 1, \dots, m$. This transformation scales the singular values of P from the interval $[0, 1)$ to the positive real line while preserving orientation.

Finally, for completion, one can recover P from A by applying the following inverse transformation:

$$P = (I_m + AA^\top)^{-\frac{1}{2}} A. \quad (27)$$

4.1.3 Prior distribution

The relationship between the singular value decompositions of P and A , discussed earlier, has important implications. The spectral norms of P and A , $r_1 = \|P\|$ and $\tilde{r}_1 = \|A\|$, are connected through the monotonic mapping $\tilde{r}_1 = r_1/(1 - r_1^2)^{1/2}$. Therefore, the relative sizes of the unconstrained parameters A_s across lags $s = 1, \dots, p$ are directly related to the relative sizes of the partial autocorrelation matrices P_s . Specifically, $P_s = 0_m$ if and only if $A = 0_m$, where 0_m is the $m \times m$ zero matrix.

By definition of partial autocorrelation matrices, for $k < p$, $P_k \neq 0_m$ and $P_{k+s} = 0_m$ for $s = 1, \dots, p - k$ if and only if $\phi_k \neq 0_m$ and $\phi_{k+s} = 0_m$ over the same range. At the same time, given the bijective mapping between A and P , the order of a $\text{VAR}_m(p)$ is $k < p$ if and only if $A_k \neq 0_m$ and $A_{k+s} = 0_m$ for $s = 1, \dots, p - k$. It follows then that, under the unconstrained reparametrization, the model of order $k < p$ is nested within that of order $k + 1$. Accordingly, adopting a shrinkage prior for $A_1, \dots, A_{p_{\max}}$ with a large $p_{\max}$ — where $p_{\max}$ defines the maximum number of lags under consideration — enables learning the lag at which the A_s matrices effectively become zero. This, in turn, informs the inference on the true order p of the process.

As discussed in Section 3, it is reasonable to assume that recent lags contribute more significantly to the prediction process than distant ones. This assumption suggests that partial autocorrelations should be higher at lower lags and diminish as the lag order increases. Using the Multiplicative Gamma Process (MGP) prior, which enforces increasing shrinkage on higher-order lags, aligns well with this assumption, making it appropriate as a prior distribution for the transformed partial autocorrelations A_s . Denoting the collection of unknown hyperparameters in the MGP by ϑ , the overall prior specification takes the following form:

$$\pi(\Sigma, A_1, \dots, A_{p_{\max}}, \vartheta) = \pi(\Sigma)\pi(\vartheta) \prod_{s=1}^{p_{\max}} \pi(A_s|\vartheta), \quad (28)$$

where the distribution of the error variance Σ must be invariant to the order of the variables in the observation vector. Heaps (2023) discusses alternative distributions that fulfill this requirement, and here, as in Binks et al. (2023), the distribution chosen is inverse Wishart.

4.1.4 Posterior inference

Denote by $\mathbf{y}_{1:n}$ the n observations of a time-series generated from a zero-mean vector autoregressive process of order $p_{\max}$, $\text{VAR}_m(p_{\max})$. Then, the corresponding likelihood is given by:

$$p(\mathbf{y}_{1:n}) = p(\mathbf{y}_{1:p_{\max}}) \prod_{t=p_{\max}+1}^n p(\mathbf{y}_t | \mathbf{y}_{(t-p_{\max}): (t-1)}). \quad (29)$$

At each time t , the conditional distribution of $\mathbf{Y}_t$ given the previous $p_{\max}$ observations $\mathbf{Y}_{(t-p_{\max}): (t-1)}$ follows a multivariate normal distribution:

$$\mathbf{Y}_t | \mathbf{Y}_{(t-p_{\max}): (t-1)}, \Sigma, \Phi \sim \mathcal{N}_m \left(\sum_{i=1}^{p_{\max}} \phi_i \mathbf{y}_{t-i}, \Sigma \right). \quad (30)$$

Besides, the initial distribution of $\mathbf{Y}_1^\top, \dots, \mathbf{Y}_{p_{\max}}^\top$ conditional on Σ and Φ is given by:

$$\mathbf{Y}_1^\top, \dots, \mathbf{Y}_{p_{\max}}^\top | \Sigma, \Phi \sim \mathcal{N}_{mp_{\max}}(0, G). \quad (31)$$

Above, the covariance matrix G is given by:

$$G = \begin{pmatrix} \Gamma_0 & \Gamma_1 & \dots & \Gamma_{p_{\max}-1} \\ \Gamma_1^\top & \Gamma_0 & \dots & \Gamma_{p_{\max}-2} \\ \vdots & \vdots & \ddots & \vdots \\ \Gamma_{p_{\max}-1}^\top & \Gamma_{p_{\max}-2}^\top & \dots & \Gamma_0 \end{pmatrix}, \quad (32)$$

where the values $\Gamma_0, \dots, \Gamma_{p_{\max}-1}$ are by-products of the recursive mapping between the partial autocorrelations and the original model parameters.

Combining the likelihood with the prior for the parameters $\Sigma, A_1, \dots, A_{p_{\max}}$, and ϑ from Equation 28, Bayes' theorem gives the posterior distribution:

$$\pi(\Sigma, A_1, \dots, A_{p_{\max}}, \vartheta | \mathbf{y}_{1:n}) \propto \pi(\Sigma) \pi(\vartheta) \prod_{s=1}^{p_{\max}} \pi(A_s | \vartheta) p(\mathbf{y}_{1:n} | \Sigma, A_1, \dots, A_{p_{\max}}). \quad (33)$$

The posterior distribution above is a complex function of the A_s . Thus, performing computational inference using Gibbs sampling is impossible. Binks et al. (2023) suggest instead using Hamiltonian Monte Carlo (HMC), which uses information on the slope of the logarithm of the posterior density to generate global proposals for all parameters simultaneously.

Because the MGP prior does not assign probability mass at zero, none of the A_s values, and consequently none of the P_s values, shrink to zero. Hence, it is necessary to establish a criterion that determines when a value is sufficiently close to zero to be considered negligible.

Given this predefined threshold, the effective order p^* of the model is the largest $s \leq p_{\max}$ such that P_s surpasses the threshold, but P_{s+k} for any $k = 1, \dots, p_{\max}$ does not. Note that applying the truncation criterion to the standardized P_s matrices, rather than the unconstrained A_s matrices, allows classical univariate time-series theory to inform the analysis in a way that remains robust to the scale of the data, its length, and its dimension m .

Binks et al. (2023), following Bhattacharya and Dunson (2011), applies a truncation criterion where P_s is set to a zero matrix if the absolute value of all elements is below a threshold ε . In this context, ε is defined as the quantile $q_m(\beta)$ of the distribution of the maximum absolute value of the m^2 elements $p_{s,ij}$ under the assumption that they are approximately independent and normally distributed as $N(0, 1/n)$. Specifically, the threshold ε is given by:

$$\varepsilon = q_m(\beta) = \frac{1}{\sqrt{n}} \Phi^{-1} \left(\frac{\beta^{1/m^2} + 1}{2} \right), \quad (34)$$

where Φ^{-1} is the inverse cumulative distribution function of the standard normal distribution, m is the dimension, n is the sample size, and β is large, like 0.99.

4.2 Prior Distribution of the Effective Lag Order

4.2.1 Determinants of the effective lag order

The effective lag order p^* in a $\text{VAR}_m(p)$ model represents the estimated number of lags that contribute meaningfully to the dynamics of the time series. Note that within the inference framework described previously, there is no direct parametrization of the effective lag order; instead, p^* emerges as a result of the shrinkage imposed on the partial autocorrelation matrices P_s (or their transformed counterparts, the A_s -matrices) across different lags. This indirect relationship means imposing a clear, explicit prior on p^* is difficult.

Based on the Bayesian inference framework described in Section 4.1, two primary factors influence the determination of p^* . First, the Multiplicative Gamma Process (MGP) prior applied to the A_s -matrices to control the shrinkage of lag coefficients. Second, the truncation threshold ε that determines when to discard small partial autocorrelations P_s .

The MGP prior affects the shrinkage of lag coefficients by progressively shrinking higher-order lags more aggressively as the lag order increases. The global shrinkage parameters $\tau[s]$, determined by the hyperparameters a_1 and a_2 , drive the partial autocorrelations P_s toward zero, allowing the model to infer when additional lags no longer contribute meaningfully to the time series dynamics. While the maximum lag order $p_{\max}$ sets the upper limit on the number of lags considered, it does not directly control the effective lag order p^* . Be-

yond a certain point, increasing $p_{\max}$ further has little effect because the shrinkage process penalizes higher-order lags, stabilizing p^* as the model naturally forces these coefficients toward zero.

The second factor is the truncation threshold ε , which determines when partial autocorrelations P_s are effectively zero. Three variables determine the threshold, that is, the sample size n , dimensionality m , and the probability parameter β . Increasing the sample size n while holding β and m constant lowers the threshold, making it harder to truncate lags. Conversely, increasing the dimensionality m or β raises the threshold, making truncation easier. Note that while the truncation threshold adapts based on sample size and dimensionality, it remains independent of $p_{\max}$, meaning that increasing the level of overparametrization does not affect the truncation process.

4.2.2 Algorithm for generating prior distribution of the effective lag order

To better understand the behavior of the prior of the effective lag order p^* , we devised a Monte Carlo algorithm to sample from the prior distribution of p^* . This algorithm generates the global shrinkage parameters τ_s for each lag and local precision parameters $\lambda_{s,i,j}$, and using those, the $a_{s,i,j}$ according to the specification of the MGP prior described in Section 3.1. It then recovers the partial autocorrelation matrices from the A_s matrices using the bijective mapping defined in Equation 27. Finally, using the truncation threshold for pre-defined levels of m , n , and β , it is possible to obtain the effective lag order for that sample — i.e., one drawing of p^* .

The algorithm can also toggle between the original MGP or its extended version (Section 3.2), depending on the boolean `if_original`. In case `if_original` is false, the local precision parameter rate is random and follows a gamma distribution. Otherwise, it is constant and set to the same as the shape parameter. Following is the pseudo-code describing how to obtain draws of the prior for p^* :

Algorithm 1 Generate Priors

```

1: Input:  $samples, m, p_{\max}, N, \beta, a_1, a_2, c_1, c_2, a, \text{if\_original}$ 
2: Output:  $optimal\_p$ 
3:  $optimal\_p \leftarrow []$ 
4: for  $i = 1$  to  $samples$  do
5:   Generate  $\tau$  (global shrinkage):
6:    $\tau[1] \leftarrow \text{RGamma}(a_1, 1)$ 
7:   for  $s = 2$  to  $p_{\max}$  do
8:      $\tau[s] \leftarrow \tau[s-1] \times \text{RGamma}(a_2, 1)$ 
9:   end for
10:  Generate  $\lambda$  (local shrinkage) and  $A$  matrices:
11:  for  $s = 1$  to  $p_{\max}$  do
12:    for  $i = 1$  to  $m$  do
13:      for  $j = 1$  to  $m$  do
14:        if  $\text{if\_original}$  then
15:           $\eta \leftarrow a$ 
16:        else
17:           $\eta \leftarrow \text{RGamma}(c_1, c_2)$ 
18:        end if
19:         $\lambda[i, j, s] \leftarrow \text{RGamma}(a/2, \eta)$ 
20:         $\text{variance} \leftarrow \frac{1}{\lambda[i, j, s] \times \tau[s]}$ 
21:         $A\_matrices[i, j, s] \leftarrow \text{RNormal}(0, \sqrt{\text{variance}})$ 
22:      end for
23:    end for
24:  end for
25:  Convert  $A$  to  $P$  matrices:
26:  for  $s = 1$  to  $p_{\max}$  do
27:     $C \leftarrow \text{IdentityMatrix}(m) + A\_matrices[:, :, s] \times \text{Transpose}(A\_matrices[:, :, s])$ 
28:     $B \leftarrow \text{SqrtMatrix}(C)$ 
29:     $P\_matrices[:, :, s] \leftarrow \text{Inverse}(B) \times A\_matrices[:, :, s]$ 
30:  end for
31:  Generate threshold for truncation:
32:   $\text{threshold} \leftarrow \frac{\text{QNorm}\left(\left(\beta \frac{1}{m^2} + 1\right)/2\right)}{\sqrt{N}}$ 
33:  Compute optimal  $p$ :
34:   $optimal\_p\_value \leftarrow 0$ 
35:  for  $s = 1$  to  $p_{\max}$  do
36:    if  $\text{MaxAbsValue}(P\_matrices[:, :, s]) \leq \text{threshold}$  then
37:      break
38:    else
39:       $optimal\_p\_value \leftarrow s$ 
40:    end if
41:  end for
42:   $optimal\_p \leftarrow \text{Append}(optimal\_p\_value, optimal\_p)$ 
43: end for
44: Return:  $optimal\_p$ 

```

4.3 Simulation Design and Performance Evaluation

4.3.1 Evaluating the prior distribution of the effective lag order

This section outlines the procedure for evaluating the extended prior distribution from Section 3.2, where the rate hyperparameter of the local precision parameters is random, compared to the specification from Section 3.1, which has a constant rate set equal to the shape hyperparameter. The main goal is to examine how different hyperparameter configurations influence the prior distribution of the effective lag order, p^* , especially considering the degree of shrinkage. Evaluating this involves generating histograms of p^* using Algorithm 1 under various hyperparameter settings and comparing the results through graphical inspection.

For the original prior specification from Section 3.1, two different hyperparameter configurations are considered. The first, with $a_1 = 3$ and $a_2 = 2.5$, as in Binks et al. (2023), induces stronger shrinkage. In contrast, the second configuration, with $a_1 = a_2 = 2$, is more flexible, allowing for less aggressive global shrinkage according to Durante (2017). For the extended prior, the evaluation is conducted with $a_1 = a_2 = 2$ while varying the hyperparameters c_1 and c_2 . All remaining hyperparameters are kept constant across the different settings to ensure consistency in the comparison.

The settings specified above are evaluated for all combinations of $p = 10, 20, \dots, 100$ and $m = 1, 3, 5$, for both the original and extended priors. For each combination, 5000 independent observations of the effective lag order p^* are drawn. The complete list of the hyperparameter combinations follows in Table 1:

Table 1: Hyperparameter settings for each scenario

Hyperparameter	Original	Original (updated)	Extended
a1	2.5	2	2
a2	3	2	2
(c1, c2)	-	-	[(1, 1), (10, 0.1), (100, 0.01), (1000, 0.001), (10000, 0.0001)]

Note: Other hyperparameters consistent across all scenarios; $a = 6$, $\beta = 0.99$, samples = 5000, and $N = 10000$. The $\text{VAR}_m(p_{\max})$ processes considered include all possible combinations of $m = [1, 3, 5]$ and $p_{\max} = 10, 20, \dots, 100$ for all scenarios.

This analysis will focus on several key aspects in comparing the original and extended priors. For the original prior, the primary objective is to identify the level of overparameterization at which the distribution converges, indicating where further lags no longer receive meaningful

mass. This point marks the maximum level of overparameterization suitable for inference, as beyond it, lags are not assigned significant mass in either the prior or posterior distributions. Meanwhile, for the more flexible specification, the assessment is how they enhance the adaptability of the p^* prior distribution.

In addition to the considerations above, the framework will consider the role of nested Gamma distributions in the extended specification. These distributions will offer insights into how the size of the draws of the local precision parameters $\tau_{s,i,j}$ behave across different c_1 and c_2 configurations. As for the prior distribution of the effective lag order, this analysis will rely on a graphical inspection of histograms.

4.3.2 Simulating data from a stationary VAR process

The process of simulating data from a stationary $\text{VAR}_m(p)$ process begins by generating the set of unconstrained matrices A_s , where each element of A_s is drawn independently from a standard normal distribution. Subsequently, the A_s matrices are transformed into partial autocorrelation matrices P_s using the bijective mapping described in Equation 25. A recursive algorithm then transforms the partial autocorrelation matrices P_s into the autoregressive coefficient matrices ϕ_s and auxiliary matrices Γ_s . A stationarity check then ensures that all eigenvalues of the $mp \times mp$ coefficient matrix have absolute values less than one.

After initialization of the autoregressive coefficients, the time series sample follows from an iterative procedure. First, the process initializes the initial p observations of the time series using a normal distribution with mean μ — set to a vector of zeros — and covariance matrix G . Following this initialization, each subsequent observation is generated iteratively based on the past p observations. Specifically, for each time step $t > p$, the new observation y_t is computed as a linear combination of the past p observations, adjusted by the autoregressive coefficients ϕ , and centered around the mean μ . The recursion is given by:

$$y_t = \mu + \sum_{i=1}^p \phi_i(y_{t-i} - \mu) + \epsilon_t, \quad (35)$$

where ϵ_t is a random error term sampled from a multivariate normal distribution with mean zero and covariance matrix Σ . This process repeats for each time step until the desired length of the time series — N . A full description of the procedure, which includes an explanation of the construction of matrix G , is available on Heaps (2023).

4.3.3 Comparing the impact of prior specifications on lag inference accuracy

This section describes the methodology for evaluating how different priors for the transformed partial autocorrelations A_s impact the accuracy of lag order inference. First, using the procedure outlined in Section 4.3.2, data from various $VAR_m(p)$ models are generated. For each combination of m and p considered, and across multiple experiments, the Bayesian inference framework from Section 4.1 determines the effective lag order. The hit rates of each specification serve as the primary performance metric, providing a direct measure of how accurately each prior identifies the correct lag order. Additionally, the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) hit rates serve as performance benchmarks.

Binks et al. (2023) analyzes datasets generated from VAR processes with smaller lag orders, specifically $p = 1, 2, 3, 4$ across dimensions $m = 1, 3, 5, 7$. While the hit rate for those orders is high, simulations for larger lag orders — particularly those requiring high overparametrizations — reveal a significant degradation in performance. To illustrate this, simulation results for data generated from a VAR process with low and moderate lag orders $p = 1, 3, 7, 12$ and 30 and single dimension $m = 1$ are presented. The accuracy of this original inference framework is then compared to that using global shrinkage parameters $a_1 = a_2 = 2$ (see Section 4.3.1) and extended MGP prior for the transformed partial autocorrelation (see Section 3.2) for data generated from a VAR process of a single dimension and $p = 12$ lags.

The simulation conducts five experiments for each combination of m , p , and prior specification, using the median of the posterior distribution to determine the effective lag order for calculating the hit rate. In all cases employing the original MGP specification, it runs four chains, each comprising 1,000 warm-up iterations followed by 4,000 sampling iterations, with overparametrization varying to reflect the number of lags. Ideally, the simulation would use the same setup for the original specification with flexible global shrinkage and the extended specification. However, computational limitations make this very difficult. Consequently, for these cases, it runs one chain comprising 800 warm-up iterations and 2,000 sampling iterations, with an overparametrization level fixed at 20 lags.

4.3.4 Implementation

The analysis uses R, with posterior inference performed via Hamiltonian Monte Carlo (HMC) through the probabilistic programming language Stan. Stan requires users to write a program in the Stan modeling language that computes the logarithm of the kernel of the posterior density function. Once defined, Stan configures a Markov chain simulation to generate samples from the posterior distribution. This process calculates the gradient of the log-

arithmetic of the posterior density, initializes the chains randomly, and tunes the sampler to ensure efficient performance.

For this project, the `cmdstanr` package, a lightweight R interface to Stan, facilitates model fitting and sampling. The Bayesian inference framework code, including the Stan code, builds on the work of Binks et al. (2023)¹. However, their original code underwent significant modifications and extensions. For example, both the R code and Stan code now incorporate the extended multiplicative gamma process prior, and the output structure has a new configuration that supports the analyses described earlier. Additionally, the code for evaluating the effective lag order prior distribution, Markov chain validation, and all graphical output are original.

All related code, the generated datasets, and the analysis results are available on the GitHub repository.

¹The repository is available here.

5 Simulation Results

5.1 Prior Distribution of the effective lag order

5.1.1 Overview

The following four sections examine the implicit prior distribution of the effective lag order p^* . As noted earlier, p^* lacks explicit parametrization; instead, the prior distribution applies to the transformed partial autocorrelations. Through the previously described bijective mapping, these transformed partial autocorrelations convert into partial autocorrelations, with a truncation criterion determining which lag orders are deemed insignificant based on these values. Hence, the implied distribution depends on the truncation criterion and the specification of the prior distribution over the transformed partial autocorrelations. Each aspect is initially studied independently. Then, the empirical distributions of the p^* constructed using the original (Section 3.1) and extended MGP (Section 3.2) prior for the transformed partial autocorrelations follow.

5.1.2 Truncation threshold

Figure 1 displays the truncation threshold ε of Equation 34 as a function of m , N , and β . The figure includes three subplots, each illustrating how the truncation threshold changes with one parameter while holding the other two fixed. Specifically, the first subplot shows the threshold varying with m when $\beta = 0.99$ and $N = 10,000$. The second subplot depicts the variation with β while $m = 3$ and $N = 10,000$. Finally, in the third N varies, with $m = 3$ and $\beta = 0.99$:

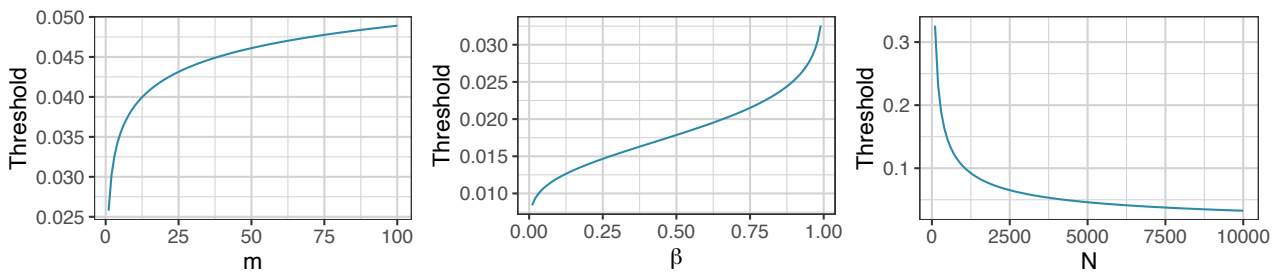


Figure 1: Threshold values as functions of parameters β , m , and N for varying conditions: (Left) Threshold versus β for $m = 3$ and $N = 10,000$; (Middle) Threshold versus m for $\beta = 0.99$ and $N = 10,000$; (Right) Threshold versus N for $\beta = 0.99$ and $m = 3$.

Looking at the first subplot, the threshold ε increases as m increases in a smooth, monotonic fashion. For small values of m , particularly $m \leq 10$, ε rises relatively steeply from around

0.025 to approximately 0.04. As m increases beyond that, the curve begins to flatten, with threshold values approaching 0.05. In the second subplot, the threshold ε also increases monotonically with β . Specifically, at higher levels of β , particularly above 0.85, the rate of increase accelerates sharply, with the threshold reaching around 0.03 as β approaches 1. Finally, converse to the other parameters, the threshold decreases monotonically with N .

5.1.3 Behavior of nested gamma distributions

As anticipated, to avoid overshrinkage of the lag order, we explore the use of a novel extension of the MGP where the gamma prior for the local shrinkage parameters $\lambda_{s,i,j}$ is substituted by a Nested Gamma prior (as defined in Equation 2). The histograms in Figures 2, 3, and 4 are constructed using 100,000 Monte Carlo samples. In each figure, the left panel shows the distribution of samples from a standard Gamma distribution with specified shape and rate parameters. Meanwhile, the right panel displays samples from the corresponding nested Gamma distribution, with an outer shape fixed at 2.5 and scale parameters derived from the inner Gamma distribution.

In Figure 2, the Gamma(10,1) distribution, with a shape parameter of 10 and a rate of 1, appears approximately symmetric, with a peak around its mean of 10. This configuration produces a shape with most values located around the mean and a moderate tail extending toward higher values with weak positive skewness. The corresponding nested Gamma distribution, Gamma[2.5, Gamma(10,1)], shows a concentration of values near zero while allowing occasional larger values in the extended tail.

Compared to the previous specification, the Gamma(100, 0.1) in Figure 3 shows a sharper peak around its mean of 1,000, with most values tightly located near the mean and a limited tail toward higher values. In contrast, the nested Gamma[2.5, Gamma(100, 0.1)] distribution results in a compact range, with values densely concentrated between 0 and 0.015. This difference highlights the effect of increasing the shape-to-rate gap in the inner Gamma distribution, leading to progressively higher concentration near zero in the nested distribution.

As the shape-to-rate gap widens further in the case of Gamma(1000, 0.01) and nested Gamma[2.5, Gamma(1000, 0.01)], these effects intensify. Notably, the nested Gamma compresses even more, with most draws between 0 and 0.00015, evident in Figure 4.

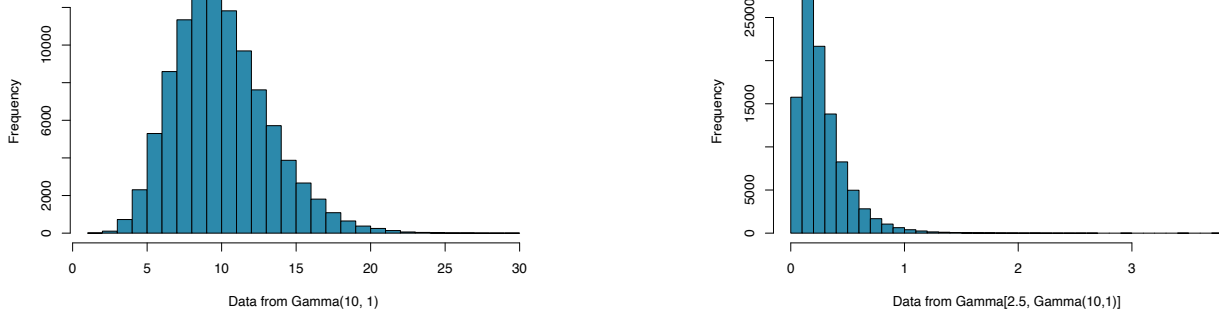


Figure 2: Histograms of (left) 100,000 Monte Carlo samples from a $\text{Gamma}(10, 1)$ distribution and (right) from a nested $\text{Gamma}[2.5, \text{Gamma}(10, 1)]$ distribution.

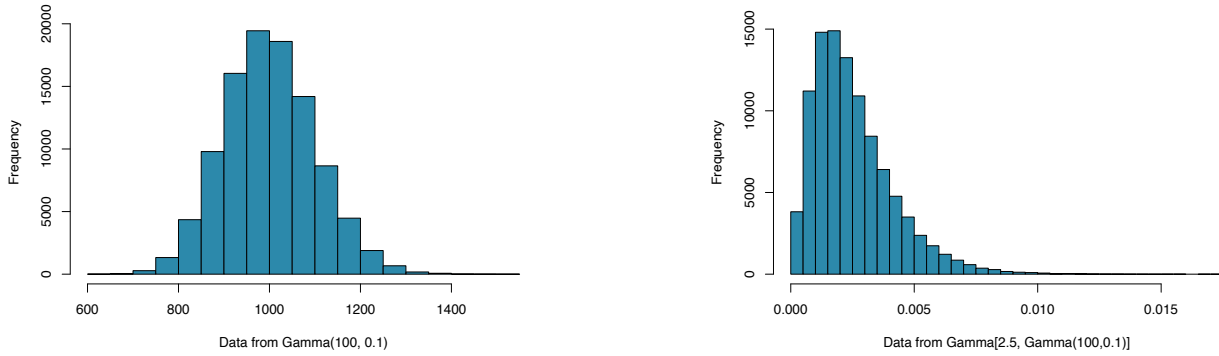


Figure 3: Histograms of (left) 100,000 Monte Carlo samples from a $\text{Gamma}(100, 0.1)$ distribution and (right) from a nested $\text{Gamma}[2.5, \text{Gamma}(100, 0.1)]$ distribution.

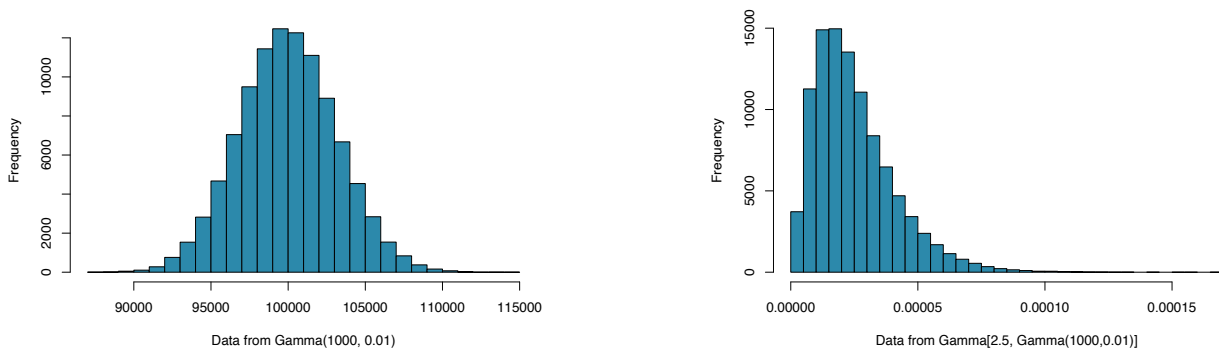


Figure 4: Histograms of (left) 100,000 Monte Carlo samples from a $\text{Gamma}(1000, 0.01)$ distribution and (right) from a nested $\text{Gamma}[2.5, \text{Gamma}(1000, 0.01)]$ distribution.

5.1.4 Prior distribution of the effective lag order (original MGP)

This section presents simulation results for the effective order p^* across various overparameterization levels and process dimensions. Figure 5 shows histograms for dimensions $m = 1, 3, 5$ and overparameterizations $p_{\max} = 6, 8, 10, 20$ and 100, using the MGP prior with global shrinkage hyperparameters $a_1 = 3$ and $a_2 = 2.5$. Figure 6 compares histograms for the same dimensions and overparameterizations $p_{\max} = 20, 30, 40, 60$ and 100 under the original global shrinkage settings ($a_1 = 3, a_2 = 2.5$, in blue) and more flexible settings ($a_1 = a_2 = 2$, in orange). All cases use $a = 6$. To further quantify the results, Table 2 presents summary statistics of the effective lag order p^* for different overparameterizations, dimensions, and global hyperparameter configurations.

Examining Figure 5, for $p_{\max} = 6$, there is a sharp concentration of the effective lag order distribution at the maximum allowed order, more prominent at higher dimensions. This suggests that truncation does not occur within the limits of $p_{\max} = 6$, possibly because the global shrinkage is too weak, leading to most of the distribution's mass concentrating around the upper boundary.

With $p_{\max} = 8$ and $p_{\max} = 10$, an observable improvement in the distribution's shape is evident across dimensions. Specifically, for $m = 1$, the distribution becomes more even by $p_{\max} = 8$, though some concentration near the upper boundary persists. By $p_{\max} = 10$, the shape improves further, with mass gradually declining after about $p^* = 6$. One possible explanation is that the truncation threshold remains invariant to the level of overparameterization. Therefore, for truncation to occur, the transformed partial autocorrelations must be sufficiently small, which—given the global shrinkage hyperparameters—requires a specific level of overparameterization.

At $p_{\max} = 20$, mass concentrates around $p^* = 5$ for $m = 1$ and around $p^* = 9$ for $m = 3$ and 5, with negligible mass beyond $p^* = 12$ across all dimensions. This pattern signals a practical limit to the distribution's shape under this level of overparameterization. With mass concentrated at or below $p^* = 12$, the posterior would similarly fail to assign meaningful mass beyond this level after Bayesian updating. Indeed, this may explain the Bayesian inference framework's limitations in capturing lag orders for processes like $\text{VAR}_m(p) = \text{VAR}_1(12)$.

Across all dimensions, when $p_{\max}$ reaches 100, the distribution shows no meaningful difference from that at $p_{\max} = 20$. The distribution continues to assign negligible mass beyond $p^* = 12$, underscoring the framework's limitation in accommodating higher effective lag orders, even with significantly increased overparameterization. Table 2 further supports this observation; for instance, for $m = 1$ and $a_1 = 2.5, a_2 = 3$, the median effective lag order remains constant at 5 across different $p_{\max}$, and the maximum observed p^* remains at 21.

According to Durante (2017), the configuration with global shrinkage hyperparameters $a_1 = a_2 = 2$ should be more flexible. Indeed, looking at Figure 6, there is less aggressive truncation across all dimensions and levels of overparameterization compared to the configuration with $a_1 = 2.5$ and $a_2 = 3$. Table 2 further substantiates this, as the median p^* increases significantly, especially at higher overparameterizations.

Despite, even under the more flexible specification, the prior distribution still fails to assign meaningful mass after a specific level of overparameterization. Specifically, across all dimensions, the distributions for $p_{\max} = 60$ and $p_{\max} = 100$ are almost the same. Table 2 further verifies this. For example, for $m = 3$ under $a_1 = a_2 = 2$, the median p^* remains at 17 or 18 from $p_{\max} = 20$ up to $p_{\max} = 100$, with the standard deviation stabilizing around 8.3 to 8.5. Accordingly, this suggests the original MGP inherently imposes too much shrinkage, highlighting the need for the more flexible, extended specification of Section 3.2.

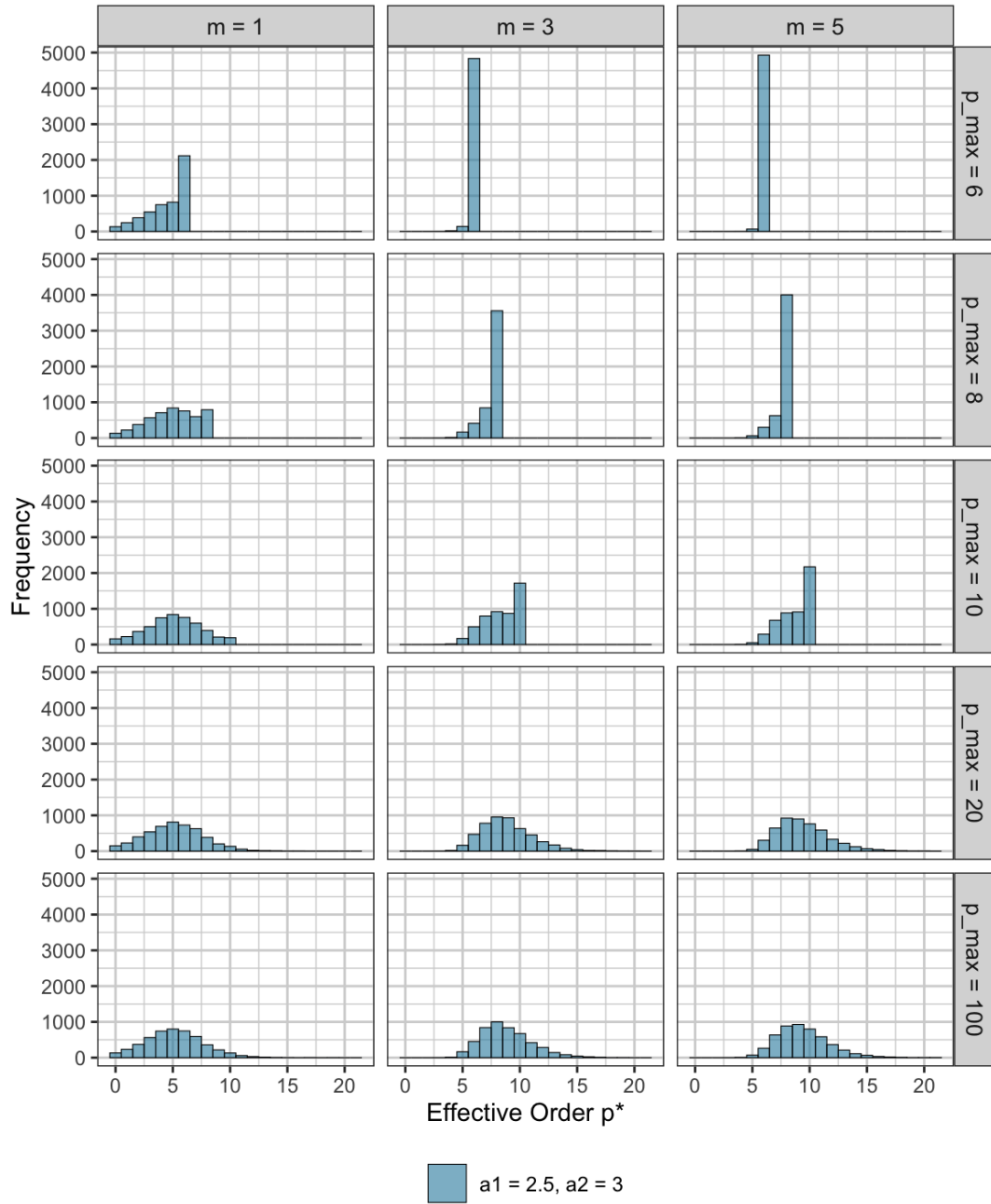


Figure 5: Histograms of 5,000 p^* samples for varying $p_{\max}$ values and $m = 1, 3$, and 5; based on the original MGP prior for the transformed partial autocorrelations with global shrinkage hyperparameters $a_1 = 2.5, a_2 = 3$.

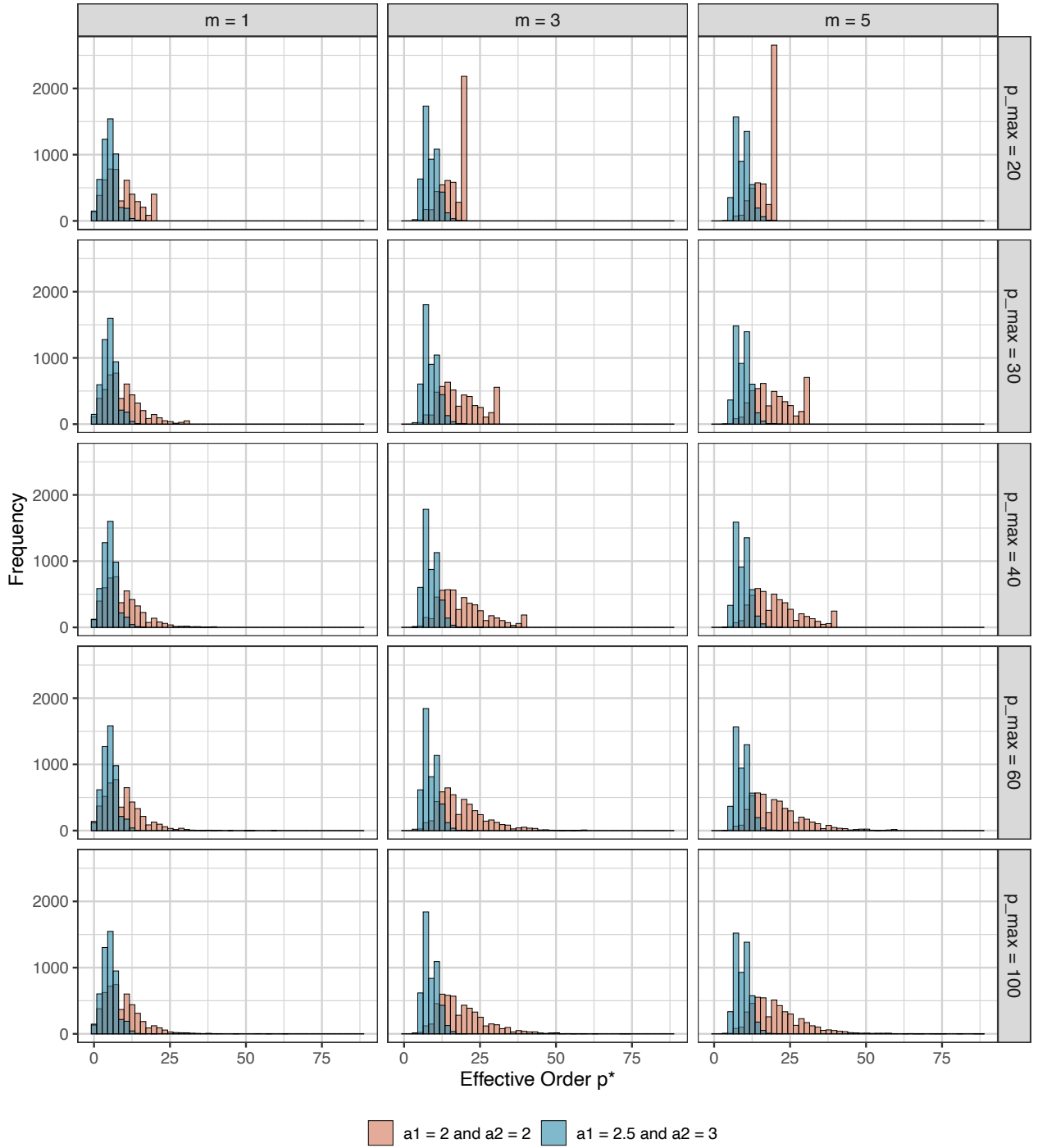


Figure 6: Histograms of 5,000 p^* samples for varying $p_{\max}$ values and $m = 1, 3$, and 5 ; based on the original MGP prior for the transformed partial autocorrelations with global shrinkage hyperparameters $a_1 = a_2 = 2$ (orange) and $a_1 = 2.5, a_2 = 3$ (blue).

Table 2: Summary statistics of effective lag order p^* for different global shrinkage hyperparameter configurations.

$a_1 = 2.5, a_2 = 3$						$a_1 = 2, a_2 = 2$					
$p_{\max}$	Min	Max	Median	SD	IQR	$p_{\max}$	Min	Max	Median	SD	IQR
$m = 1$						$m = 1$					
10	0	10	5	2.39	4	10	0	10	8	3.01	5
20	0	16	5	2.53	4	20	0	20	8	5.35	7
30	0	15	5	2.51	4	30	0	30	8	6.00	7
40	0	17	5	2.49	3	40	0	40	8	6.22	7
50	0	15	5	2.56	4	50	0	50	8	6.24	7
60	0	15	5	2.47	4	60	0	60	8	6.30	7.25
70	0	21	5	2.51	4	70	0	65	8	6.24	7
80	0	15	5	2.54	4	80	0	58	8	6.42	7
90	0	16	5	2.52	4	90	0	70	8	6.39	7
100	0	19	5	2.53	4	100	0	63	8	6.34	7
$m = 3$						$m = 3$					
10	3	10	9	1.53	3	10	5	10	10	0.53	0
20	3	20	9	2.24	3	20	5	20	17	3.95	7
30	3	20	9	2.25	3	30	5	30	17	6.62	10
40	4	25	9	2.25	3	40	4	40	18	7.94	11
50	3	21	9	2.25	3	50	4	50	17	8.39	11
60	4	21	9	2.23	3	60	5	60	17	8.20	10
70	4	20	9	2.25	3	70	5	70	18	8.49	11
80	4	21	9	2.19	3	80	5	80	18	8.46	10.25
90	3	20	9	2.27	3	90	5	72	17	8.46	10
100	3	19	9	2.21	3	100	5	73	18	8.30	10
$m = 5$						$m = 5$					
10	4	10	9	1.35	2	10	5	10	10	0.34	0
20	4	20	9	2.30	3	20	5	20	19	3.52	6
30	4	22	9	2.24	3	30	5	30	19	6.52	11
40	4	21	9	2.24	3	40	6	40	19	8.07	11
50	4	20	9	2.21	3	50	6	50	19	8.48	11
60	4	21	9	2.26	3	60	6	60	19	8.82	11
70	4	21	9	2.26	3	70	5	70	19	8.80	11
80	4	22	9	2.25	3	80	6	76	19	8.74	11
90	4	21	9	2.30	3	90	5	89	19	8.80	11
100	3	21	9	2.27	3	100	6	88	19	9.06	10

5.1.5 Prior distribution of the effective lag order (extended MGP)

This section presents the simulation results of the implicit prior distribution over effective lag order p^* under varying local shrinkage rate hyperparameters c_1 and c_2 . In the figures, blue histograms correspond to the extended MGP with flexible global shrinkage ($a_1 = a_2 = 2$), while the orange histograms correspond to the original MGP with flexible local shrinkage. For each combination of c_1 and c_2 , distributions are constructed for dimensions $m = 1, 3$, and 5 , focusing on relevant $p_{\max}$ values that capture critical ranges of overparameterization. All cases use $a = 6$.

Figure 7 illustrates the case where $c_1 = 1000$ and $c_2 = 0.001$. The histograms show that for $p_{\max} = 60$, the distribution concentrates near the maximum allowed order, indicating minimal truncation. As $p_{\max}$ increases to 80 and 100, the mass shifts away from the upper limit. For these levels of overparameterization, the median centers around 50 for $m = 3$ and $m = 5$ and around 40 for $m = 1$. Notably, across all dimensions and levels of overparameterization, the distributions allocate little mass to lower lag orders, while some mass—even if small—remains assigned to the highest orders, including 100.

These observations suggest that the combination $c_1 = 1000$ and $c_2 = 0.001$ is theoretically optimal when the prior belief anticipates a relatively high order between 30 and 60. If very high lags are possible, an overparameterization level of 100 is best; otherwise, 80 should suffice since the medians are the same, and higher overparameterizations require more computational resources.

Figures 8 and 9 present cases with smaller gaps between c_1 and c_2 , specifically $c_1 = 100, c_2 = 0.01$ and $c_1 = 10, c_2 = 0.1$. Generally, these figures show that the wider the gap between c_1 and c_2 , the stronger the ability of the additional layer of complexity to counteract global shrinkage. Indeed, according to Section 5.1.3, wider gaps lead to smaller draws of the local shrinkage parameter. Since the variance of the prior for the transformed partial autocorrelations depends inversely on these draws, this results in less overall shrinkage.

Based on this analysis, Tables 3 and 4 provide general recommendations for selecting appropriate overparameterization levels for different combinations of c_1 and c_2 .

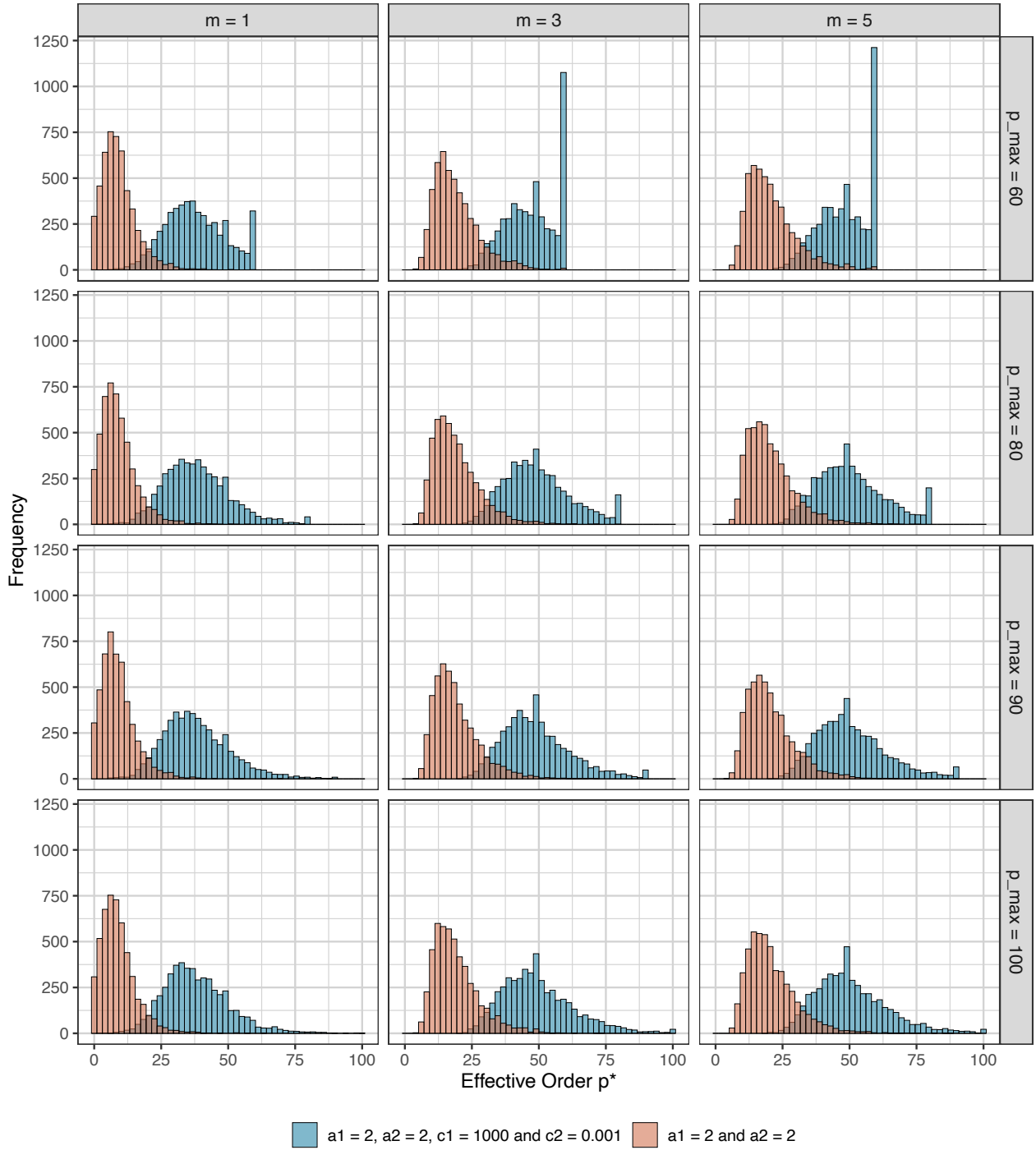


Figure 7: Histograms of 5,000 p^* samples for varying $p_{\max}$ values and $m = 1, 3$, and 5 ; based on the original MGP prior for the transformed partial autocorrelations with global shrinkage hyperparameters $a_1 = a_2 = 2$ (orange), and the extended MGP prior with $a_1 = a_2 = 2$, $c_1 = 1000$, and $c_2 = 0.001$ (blue).

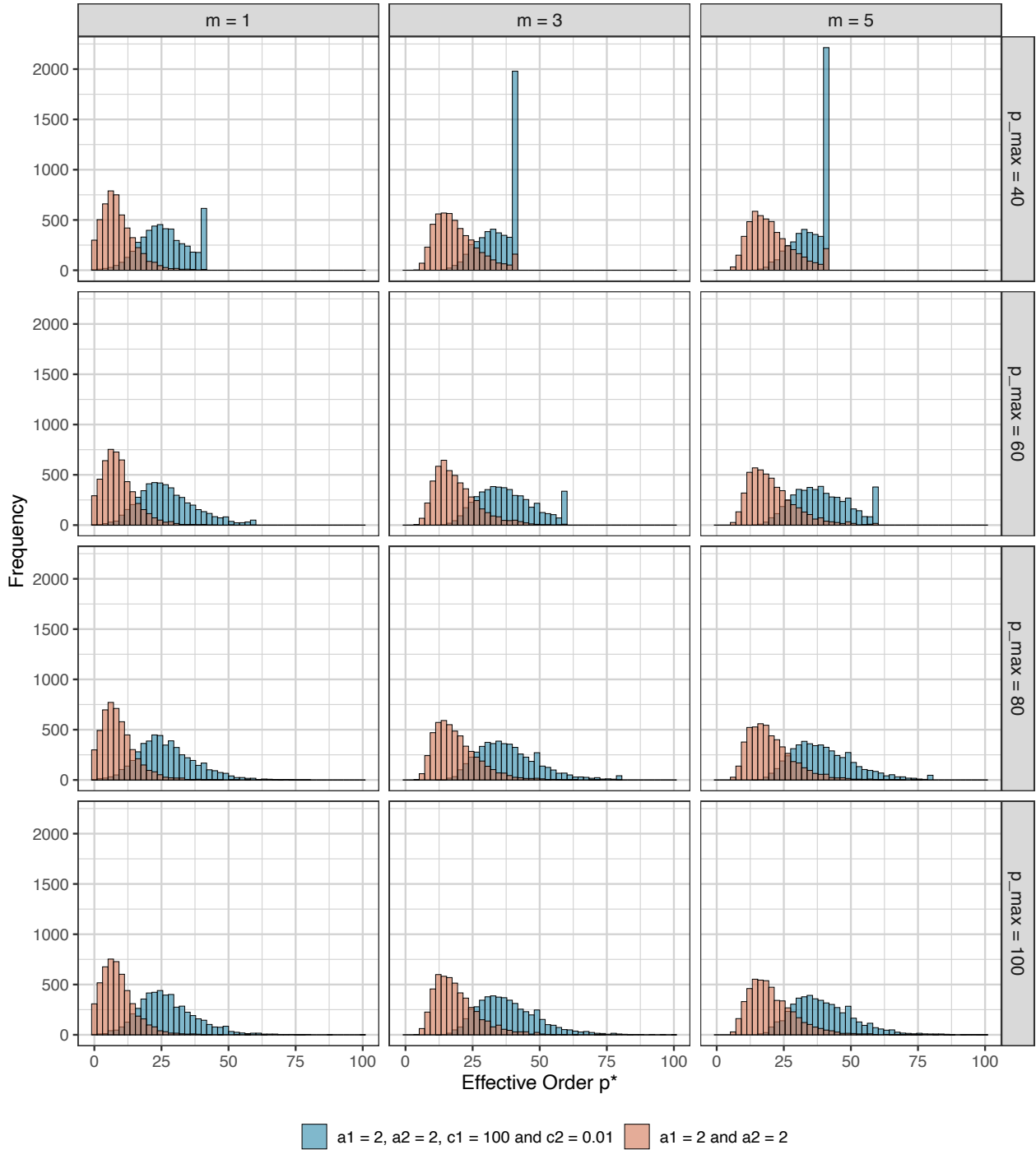


Figure 8: Histograms of 5,000 p^* samples for varying $p_{\max}$ values and $m = 1, 3$, and 5; based on the original MGP prior for the transformed partial autocorrelations with global shrinkage hyperparameters $a_1 = a_2 = 2$ (orange), and the extended MGP prior with $a_1 = a_2 = 2, c_1 = 100$ and $c_2 = 0.01$ (blue).

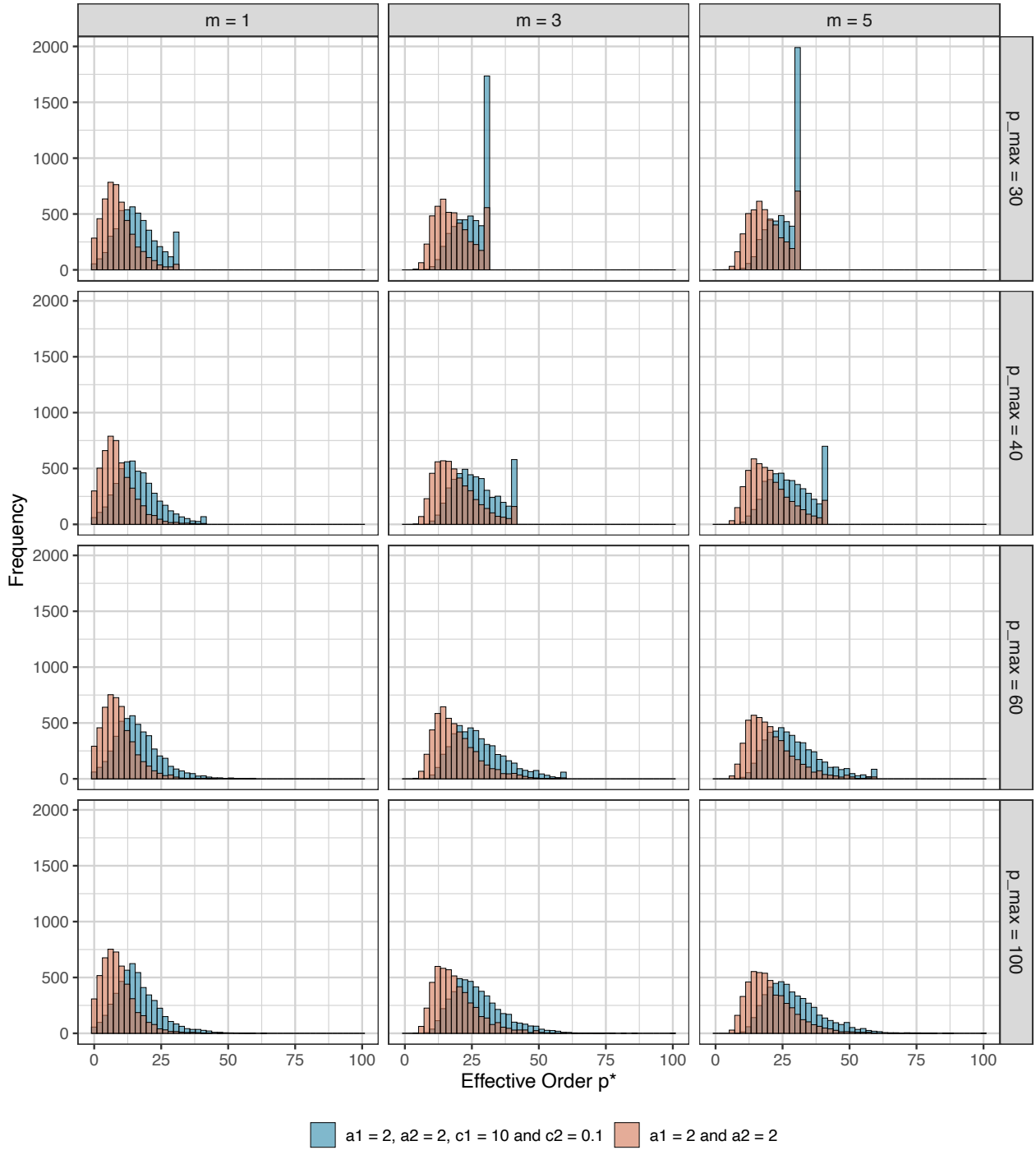


Figure 9: Histograms of 5,000 p^* samples for varying $p_{\max}$ values and $m = 1, 3$, and 5 ; based on the original MGP prior for the transformed partial autocorrelations with global shrinkage hyperparameters $a_1 = a_2 = 2$ (orange), and the extended MGP prior with $a_1 = a_2 = 2, c_1 = 10$ and $c_2 = 0.1$ (blue).

Table 3: Summary statistics for $c_1 = 5$, $c_2 = 0.5$ and $c_1 = 10$, $c_2 = 0.1$. For each m , c_1 and c_2 combination, $p_{\max}$ entries comprise most relevant overparametrization ranges.

$c_1 = 5, c_2 = 0.5$					$c_1 = 10, c_2 = 0.1$				
$p_{\max}$	Min	Max	Median	SD	$p_{\max}$	Min	Max	Median	SD
$m = 1$					$m = 1$				
20	0	20	10	5.50	40	0	40	15	7.97
30	0	30	10	6.52	50	0	50	15	8.28
40	0	40	10	7.25	60	0	60	16	8.47
$m = 3$					$m = 3$				
20	6	20	17	7.72	40	9	40	26	9.15
30	15	30	20	8.54	50	10	50	26	9.75
40	16	40	21	9.22	60	11	60	27	10.17
$m = 5$					$m = 5$				
20	6	20	17	7.72	40	19	40	27	10.23
30	17	30	22	8.54	50	21	50	27	10.82
40	17	40	23	9.13	60	22	60	27	11.03

Table 4: Summary statistics for $c_1 = 100$, $c_2 = 0.01$ and $c_1 = 1000$, $c_2 = 0.001$. For each m , c_1 and c_2 combination, $p_{\max}$ entries comprise most relevant overparametrization ranges.

$c_1 = 100, c_2 = 0.01$					$c_1 = 1000, c_2 = 0.001$				
$p_{\max}$	Min	Max	Median	SD	$p_{\max}$	Min	Max	Median	SD
$m = 1$					$m = 1$				
60	13	60	30	10.57	80	21	80	48	12.25
70	13	70	37	11.33	90	21	90	48	12.54
80	13	80	38	11.73	100	21	100	49	13.57
$m = 3$					$m = 3$				
60	15	60	30	10.95	80	22	80	49	12.35
70	18	70	38	12.25	90	23	90	49	12.73
80	19	80	38	12.54	100	24	100	50	13.10
$m = 5$					$m = 5$				
60	20	60	38	11.91	80	23	80	49	12.91
70	22	70	39	12.75	90	24	90	50	13.28
80	23	80	39	13.04	100	25	100	50	13.75

5.2 Posterior inference

5.2.1 Overview

This section presents the simulation results of the posterior distribution of the effective lag order p^* . As in the prior distribution analysis, we contrast three configurations: the original MGP specification with fixed global shrinkage hyperparameters $a_1 = 2.5$ and $a_2 = 3$, the original MGP with flexible global shrinkage, and the extended MGP specification with flexible shrinkage.

Due to computational constraints, the simulation employs different warm-up and sampling sizes, varies the number of parallel chains, and uses varying dataset sizes depending on the number of lags and prior specifications. Despite these variations, chain diagnostics using trace and autocorrelation plots confirm convergence for all chains (see Appendix B for examples). Furthermore, Appendix B provides the ϕ coefficients associated with each experiment (if available).

For the original MGP specification with global shrinkage hyperparameters $a_1 = 2.5$ and $a_2 = 3$, the simulation considers five VAR processes of a single dimension $m = 1$ and orders $p = 1, 3, 7, 12$ and 30 , each generating a dataset of length $N = 10,000$. Besides, it runs four parallel chains for the posterior distributions, each comprising 1,000 warm-up observations and 4,000 sampling observations. On the other hand, for both the original MGP with flexible global shrinkage and the extended MGP specification, the simulation considers five VAR processes of a single dimension and order $p = 12$, each generating a dataset of length $N = 500$. In these cases, it runs one chain consisting of 800 warm-up observations and 2,000 sampling observations. The setting of the additional hyperparameters in the extended specification follow from Table 3, with $c_1 = 5$ and $c_2 = 0.5$.

5.2.2 Posterior distribution (original MGP)

Table 5 and Figure 10 present the posterior distributions of the effective lag order p^* across five experiments for processes with true orders $p = 1$ and $p = 3$. Similarly, Table 6 and Figure 11 display the results for true orders $p = 7$ and $p = 12$. The bottom row of each table indicates the median of the posterior distribution, taken as the effective lag order. Additionally, Table 7 and Figure 12 extend these results to a process order of $p = 30$. For dimensions $m = 3$ and $m = 5$, similar albeit smaller studies suggest a similar pattern; however, these were omitted due to their similarity with the current results and the computational demands of conducting the full study for higher dimensions.

For processes with lower lag orders ($p = 1$ and $p = 3$), the Bayesian inference framework employing the original MGP prior with global shrinkage hyperparameters $a_1 = 2.5$ and $a_2 = 3$ demonstrates robust performance. Specifically, the framework accurately identifies the actual lag order in all but one experiment for $p = 1$, and it consistently estimates the correct order for $p = 3$ across all experiments. Figure 10 makes this clear, with most mass concentrating on the actual order for both orders.

Nevertheless, the framework's performance deteriorates as the actual lag order increases. Examination of the posterior distributions for $p = 7$ and $p = 12$ reveals a significant underestimation of the actual lag order across all experiments. For instance, in the case of $p = 12$, the estimated effective lag orders fluctuate erratically between 3 and 7 across different experiments, highlighting inconsistency and unreliability in the estimates.

This trend of declining performance becomes even more pronounced for higher-order processes, as illustrated in Table 7, which displays results for $p = 30$. Here, the estimated lag orders range from 9 to 15, with the framework failing to predict the actual lag order with a considerably higher average percentage error compared to $p = 12$.

The observed degradation in performance with increasing lag order suggests that the original MGP prior specification imposes excessive shrinkage on higher lags. Such over-shrinkage impedes the framework's ability to identify correctly the order of higher-order processes, leading to a consistent underestimation of the actual order. These findings align with the observations discussed in Section 5.1.4, where Figure 6 illustrates that for $m = 1$, the prior does not allocate meaningful mass beyond the 12th lag, regardless of overparameterization.

Table 5: Posterior distributions of the effective order p^* from five experiments. Data were generated from a $VAR_1(p)$ process with $p = 1$ lag (left) and $p = 3$ lags (right). Here, M represents the median of the posterior distribution.

$p = 1$						$p = 3$					
k	1	2	3	4	5	k	1	2	3	4	5
$m = 1$						$m = 1$					
0	0.00	0.00	0.00	0.00	0.00	0	0.00	0.00	0.00	0.00	0.00
1	0.97	0.45	0.89	0.97	0.75	1	0.00	0.00	0.00	0.00	0.00
2	0.01	0.02	0.02	0.02	0.01	2	0.00	0.00	0.00	0.00	0.00
3	0.01	0.21	0.01	0.01	0.07	3	0.97	0.85	0.69	0.95	0.79
4	0.01	0.17	0.08	0.01	0.11	4	0.01	0.14	0.27	0.01	0.20
5	0.00	0.10	0.01	0.00	0.04	5	0.01	0.01	0.04	0.00	0.01
6	0.00	0.01	0.00	0.00	0.00	6	0.00	0.00	0.02	0.00	0.00
7	0.00	0.00	0.00	0.00	0.00	7	0.00	0.00	0.00	0.00	0.00
8	0.00	0.00	0.00	0.00	0.00	8	0.00	0.00	0.00	0.01	0.00
M	1	3	1	1	1	M	3	3	3	3	3

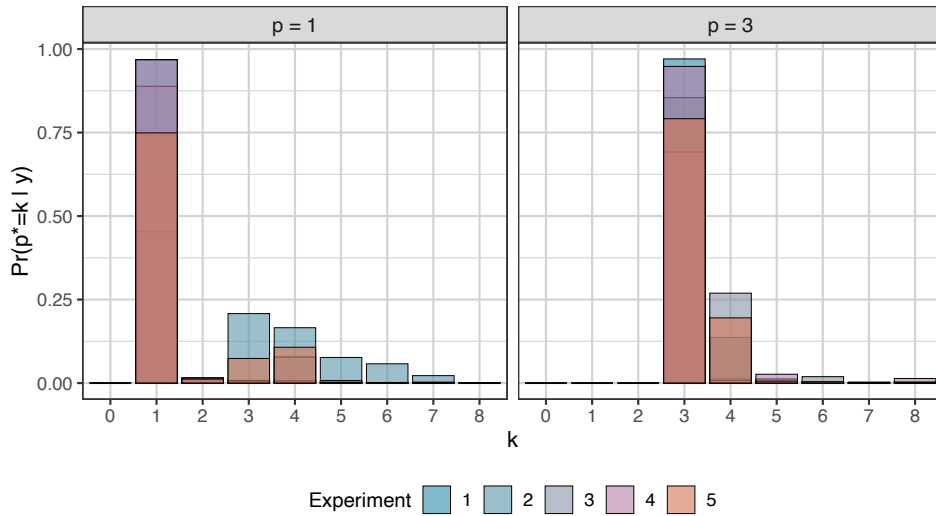


Figure 10: Overlaid posterior mass functions of the effective order p^* from five experiments with $m = 1$ and $N = 10,000$. Data were generated from a $VAR_1(p)$ process with $p = 1$ (left) and $p = 3$ (right).

Table 6: Posterior distributions of the effective order p^* from five experiments. Data were generated from a $VAR_1(p)$ process with $p = 7$ lags (left) and $p = 12$ lags (right). Here, M represents the median of the posterior distribution.

$p = 7$						$p = 12$					
k	1	2	3	4	5	k	1	2	3	4	5
$m = 1$						$m = 1$					
0	0.00	0.00	0.00	0.00	0.01	0	0.00	0.00	0.00	0.00	0.00
1	0.00	0.10	0.03	0.00	0.09	1	0.00	0.00	0.00	0.05	0.00
2	0.10	0.48	0.20	0.81	0.29	2	0.00	0.01	0.00	0.22	0.05
3	0.77	0.32	0.48	0.16	0.38	3	0.00	0.15	0.03	0.41	0.24
4	0.12	0.09	0.27	0.02	0.20	4	0.01	0.36	0.28	0.24	0.38
5	0.00	0.01	0.01	0.00	0.03	5	0.04	0.31	0.37	0.08	0.23
6	0.00	0.00	0.00	0.00	0.00	6	0.15	0.14	0.22	0.01	0.07
7	0.00	0.00	0.00	0.00	0.00	7	0.30	0.04	0.08	0.00	0.01
8	0.00	0.00	0.00	0.00	0.00	8	0.31	0.00	0.02	0.00	0.00
9	0.00	0.00	0.00	0.00	0.00	9	0.15	0.00	0.00	0.00	0.00
10	0.00	0.00	0.00	0.00	0.00	10	0.04	0.00	0.00	0.00	0.00
11	0.00	0.00	0.00	0.00	0.00	11	0.00	0.00	0.00	0.00	0.00
12	–	–	–	–	–	12	0.00	0.00	0.00	0.00	0.00
13	–	–	–	–	–	13	0.00	0.00	0.00	0.00	0.00
14	–	–	–	–	–	14	0.00	0.00	0.00	0.00	0.00
15	–	–	–	–	–	15	0.00	0.00	0.00	0.00	0.00
16	–	–	–	–	–	16	0.00	0.00	0.00	0.00	0.00
M	3	2	3	2	3	M	7	4	5	3	4

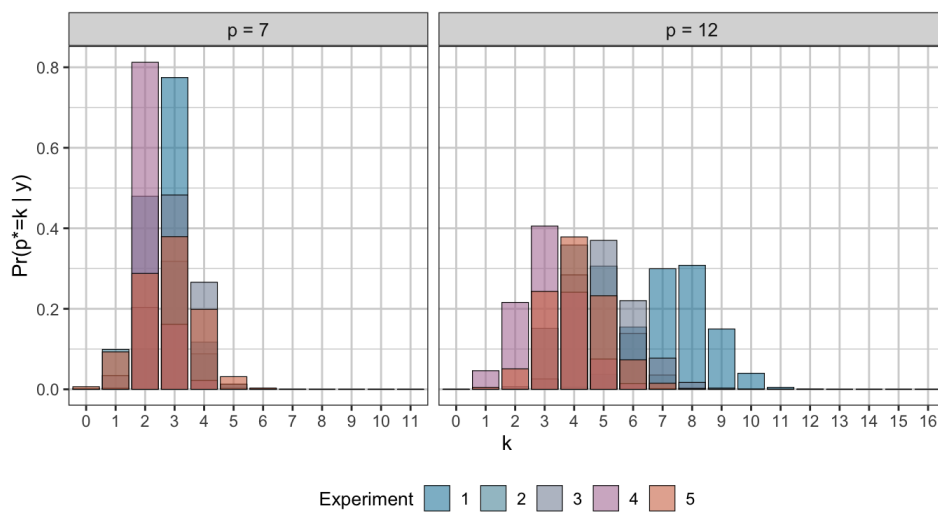


Figure 11: Overlaid posterior mass functions of the effective order p^* from five experiments with $m = 1$ and $N = 10,000$. Data were generated from a $VAR_1(p)$ process with $p = 7$ (left) and $p = 12$ (right).

5.2.3 Posterior distribution (extended MGP)

Table 8 displays the posterior distributions for the effective lag order p^* across five experiments for processes of order $p = 12$. The left panel shows the distributions constructed using the original MGP specification with flexible global shrinkage hyperparameters $a_1 = a_2 = 2$, and the right using the extended MGP also with flexible global shrinkage hyperparameters. Moreover, the last three rows display the median of the posterior distribution, the lag order implied from the AIC, and the lag order implied from the BIC. Finally, figure 5.2.3 shows the overlaid histograms of the posterior distributions for each specification.

Converse to the original MGP specification, all methods, including the AIC and BIC, accurately identify the actual lag order across all experiments. The results for the AIC and BIC are unsurprising since the order is moderate (Franses & van Dijk, 2014). In addition, a possible explanation for why the original specification with flexible global shrinkage parameters performed equally well as the extended specification is that the lag order $p = 12$ is still too low to observe the gains of the novel specification. Indeed, the simulations of the prior distribution for both specifications show that each can comfortably accommodate 12 lags at an overparametrization level of 20. However, had the simulation been conducted for a process with $p = 80$, the original specification would no longer be capable of accommodating this order since it would fall outside the range of the prior distribution (refer to Table 2). Meanwhile, it would fall within the range of the novel specification for specific parametrizations (refer to Table 4).

Due to the high computational costs associated with sampling from the posterior distribution for processes with large lag orders, achieving reliable results for high-order processes within the constraints of this work is unfeasible. Small-scale studies — i.e., generated with limited observations and posterior samples — indicate performance degradation at higher orders with the original specification, regardless of global shrinkage settings. These preliminary studies were omitted as they lacked the quality necessary for robust conclusions, serving only to tentatively corroborate the observed trend of underperformance in high-lag settings as mentioned previously.

Table 8: Posterior distributions of the effective order p^* from five experiments. Data were generated from a $VAR_1(p)$ process with $p = 12$ lags, using the original MGP (left) and the extended MGP (right) with $a_1 = a_2 = 2$. Here, M represents the median of the posterior distribution, while AIC and BIC indicate the lag orders identified by each criterion.

Original MGP with $a_1 = a_2 = 2$						Extended MGP with $a_1 = a_2 = 2$					
k	1	2	3	4	5	k	1	2	3	4	5
$m = 1$						$m = 1$					
0	0.00	0.00	0.00	0.00	0.00	0	0.00	0.00	0.00	0.00	0.00
1	0.00	0.00	0.00	0.00	0.00	1	0.00	0.00	0.00	0.00	0.00
2	0.00	0.00	0.00	0.00	0.00	2	0.00	0.00	0.00	0.00	0.00
3	0.00	0.00	0.00	0.00	0.00	3	0.00	0.00	0.00	0.00	0.00
4	0.00	0.00	0.00	0.00	0.00	4	0.00	0.00	0.00	0.00	0.00
5	0.00	0.00	0.00	0.00	0.00	5	0.00	0.00	0.00	0.00	0.00
6	0.00	0.00	0.00	0.00	0.00	6	0.00	0.00	0.00	0.00	0.00
7	0.00	0.00	0.00	0.00	0.00	7	0.00	0.00	0.00	0.00	0.00
8	0.00	0.00	0.00	0.00	0.00	8	0.00	0.00	0.00	0.00	0.00
9	0.00	0.00	0.00	0.00	0.00	9	0.00	0.00	0.00	0.00	0.00
10	0.00	0.00	0.00	0.00	0.00	10	0.00	0.00	0.00	0.00	0.00
11	0.00	0.00	0.00	0.00	0.00	11	0.00	0.00	0.00	0.00	0.00
12	0.77	0.79	0.89	0.83	0.92	12	0.85	0.82	0.89	0.83	0.89
13	0.04	0.03	0.04	0.05	0.02	13	0.02	0.03	0.03	0.04	0.02
14	0.08	0.01	0.01	0.01	0.01	14	0.08	0.01	0.01	0.00	0.01
15	0.01	0.02	0.01	0.01	0.00	15	0.01	0.01	0.01	0.01	0.00
16	0.00	0.12	0.05	0.05	0.03	16	0.00	0.11	0.05	0.06	0.05
17	0.01	0.03	0.00	0.00	0.00	17	0.00	0.03	0.00	0.00	0.00
18	0.00	0.00	0.00	0.03	0.01	18	0.00	0.00	0.00	0.03	0.01
19	0.08	0.00	0.00	0.03	0.02	19	0.04	0.00	0.00	0.03	0.02
20	0.01	0.00	0.00	0.00	0.00	20	0.00	0.00	0.00	0.00	0.00
M	12	12	12	12	12	M	12	12	12	12	12
AIC	12	12	12	12	12	AIC	12	12	12	12	12
BIC	12	12	12	12	12	BIC	12	12	12	12	12

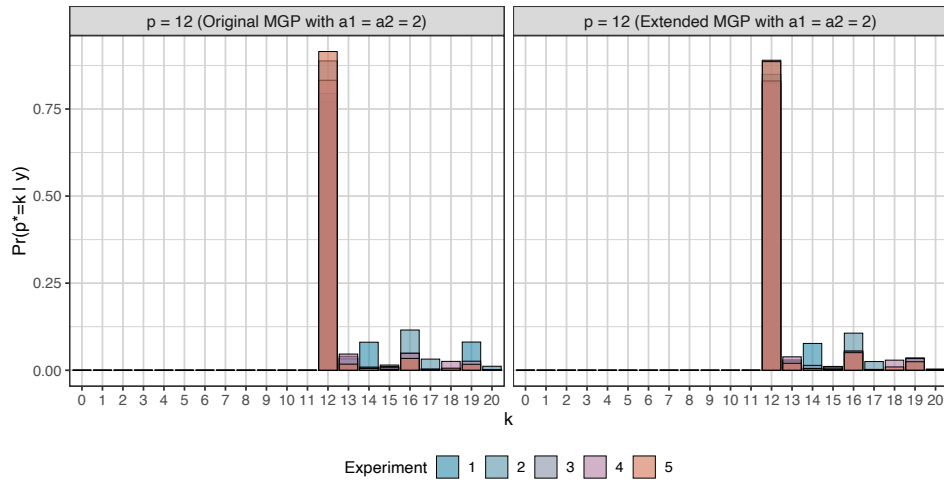


Figure 13: Overlaid posterior mass functions of the effective order p^* from five experiments with $m = 1$ and $N = 500$, using data generated from a $VAR_1(p)$ process with $p = 12$ lags. Histograms were constructed with the original MGP with flexible global shrinkage (left) and the extended MGP with flexible global shrinkage (right).

6 Discussion

6.1 Analysis of Findings

Vector autoregression (VAR) models are used widely in economics; however, lag order selection often relies on arbitrary justifications rather than theoretical rigor. Binks et al. (2023) introduce a pioneering methodology that employs a Bayesian inference framework for determining the order of stationary VARs, thereby mitigating the inherent subjectivity in existing selection methods. While this approach offers significant advantages, its performance diminishes when applied to models with large lag orders. Motivated by this limitation, the presented thesis seeks to enhance the original methodology.

The original methodology fundamentally hinges on a reparameterization of the vector autoregressive coefficients. For a vector autoregressive process to be stationary, its parameters must lie in the stationary region, a subset of the parameter space characterized by intricate and nonlinear constraints. This complexity poses significant challenges in defining appropriate prior distributions directly to the coefficients. Binks et al. (2023) circumvents this issue by reparametrizing the constrained parameters into an unconstrained Euclidean space, allowing for routine prior specification and, thereby, the construction of an inferential framework based on Bayesian principles.

Given that the method defines the prior distribution over the reparametrized coefficients, the prior distribution over the effective lag order emerges implicitly. Therefore, this thesis develops a Monte Carlo algorithm capable of sampling from this implicit prior distribution, aiming to understand the relationship between the prior distribution and algorithm performance. Simulation results corroborated the hypothesis that the inferential framework from Binks et al. (2023) relies on a prior specification that leads to excessive shrinkage and, consequently, consistent underestimation of the actual order process for large orders.

Binks et al. (2023) opted for a Multiplicative Gamma Process prior distribution with global shrinkage parameters $a_1 = 2.5$ and $a_2 = 3$, a natural choice since it imposes increasing shrinkage on later lags. Durante (2017) offers guidelines on how the choice of global shrinkage parameters affects shrinkage, and reviewing those guidelines indicated that setting $a_1 = a_2 = 2$ should provide the most flexibility while retaining the desired increasing shrinkage profile. Notwithstanding, simulating from the prior distribution suggested that, while indeed more flexible, this specification still led to overshrinkage for very high orders.

This thesis, aiming to address the previous limitations directly, proposed an extension of the Multiplicative Gamma Process that imposes a prior gamma distribution over the rate

hyperparameter of the local precision parameters, which, under the original specification, are fixed and equal to the shape hyperparameter. By introducing this additional layer of complexity, the novel specification enables the model to adjust shrinkage rates across local precision parameters according to the data's specific characteristics. Accordingly, this should allow for more tailored regularization than the fixed-rate shrinkage approach, theoretically improving performance.

Trying to test whether the novel specification of the prior distribution resulted in better performance, the thesis compared the performance of the novel specification against that of the original in selecting the actual number of lags of a process with a single dimension and 12 lags across five experiments. The original specification with Binks et al. (2023) hyperparameters underestimated the actual order for all experiments. Meanwhile, the original specification with Durante (2017) hyperparameters and the novel specification selected the correct order every time.

The original specification with Durante (2017) global shrinkage parameters may perform comparably to the extended specification because the lag order $p = 12$ is still too low to reveal the full benefits of the novel approach. Simulation studies of the prior distribution for both specifications indicate that each can effectively accommodate 12 lags. However, they also suggest that the original specification imposes excessive shrinkage at very high orders, such as $p = 80$, whereas the novel specification can comfortably handle such higher lag orders. Hence, this suggests that in settings with significantly higher lag requirements, the novel specification would effectively demonstrate its strength, positioning it as a promising tool for future applications that require accommodating higher lag orders.

6.2 Limitations and Future Research

Due to the high computational costs associated with sampling from the posterior distribution for processes with large lag orders, achieving reliable results for high orders within the constraints of this work was unfeasible. Ideally, constructing a posterior distribution for processes simulated with much higher lag values would have helped substantiate the conjecture that the novel specification of the Multiplicative Gamma Process provides performance gains over the original with appropriate hyperparameter tuning. Regardless, this presents a valuable avenue for future research, where exploring higher-lag scenarios could further validate the benefits of the extended MGP framework in large-scale VAR applications.

Another limitation of this work is the discrepancy in dataset sizes used to construct posterior distributions for the effective lag order with the original and extended Multiplicative Gamma Process priors. Evaluating both models on identical dataset sizes would have been

preferable; however, even for moderate lag orders like $p = 12$, the computational costs were prohibitive. Despite this constraint, simulations with larger sample sizes using the original MGP indicate that it may face challenges in VAR models with higher sample sizes—a scenario common when exploring higher-order VARs. This observation points to a valuable direction for future research: examining the performance of the MGP under large N and high-lag conditions despite the significant computational demands involved.

Finally, it is crucial to note that the simulation studies of the posterior distribution are limited to dimension $m = 1$, representing a simple autoregressive process rather than a full Vector Autoregression (VAR) model. This limitation, driven by the substantial computational demands of simulating higher-dimensional settings, means that the results are tentative when considering their applicability to more complex multivariate contexts—precisely where this methodology should offer the most gains over existing approaches. Extending the analysis to higher dimensions remains an essential step for future research to confirm the robustness and generalizability of the findings.

A Stan Code for Multiplicative Gamma Process Prior

```
1 functions {
2   matrix sqrtm(matrix A) {
3     int m = cols(A);
4     vector[m] sq_eigenvals = sqrt(eigenvalues_sym(A));
5     matrix[m, m] eigenvecs = eigenvectors_sym(A);
6     return mdivide_right(diag_post_multiply(eigenvecs, sq_eigenvals),
7       eigenvecs);
8   }
9   matrix A_to_P(matrix A) {
10    int m = cols(A);
11    matrix[m, m] C = diag_matrix(rep_vector(1, m)) + tcrossprod(A);
12    return mdivide_left_spd(sqrtm(C), A);
13  }
14
15  array[,] matrix P_to_AR(array[] matrix P, matrix Sigma) {
16    int p = size(P);
17    int m = cols(Sigma);
18    array[p + 1] matrix[m, m] Sigma_s, S_s, Sigma_star_s, S_star_s,
19      Gamma;
20    array[2, p] matrix[m, m] phiGamma;
21    array[p] matrix[m, m] phi_plusone, phi_plusone_star;
22    Sigma_s[p + 1] = Sigma;
23    S_s[p + 1] = sqrtm(Sigma_s[p + 1]);
24
25    for (int s in 0 : (p - 1)) {
26      matrix[m, m] Y = diag_matrix(rep_vector(1, m)) - tcrossprod(P[p -
27        s]);
28      matrix[m, m] sqY = sqrtm(Y);
29      S_s[p - s] = mdivide_right_spd(mdivide_left_spd(
30        sqY, sqrtm(quad_form_sym(Sigma_s[p - s + 1],
31          sqY))),
32        sqY);
33      Sigma_s[p - s] = tcrossprod(S_s[p - s]);
34    }
35
36    Gamma[1] = Sigma_s[1];
37    Sigma_star_s[1] = Gamma[1];
38    S_star_s[1] = S_s[1];
39    phi_plusone[1] = mdivide_right(S_s[1] * P[1], S_star_s[1]);
40    phi_plusone_star[1] = mdivide_right(S_star_s[1] * P[1]', S_s[1]);
41    Sigma_s[2] = Sigma_s[1] - quad_form_sym(Sigma_star_s[1],
```

```

phi_splusone[1]');
39   Sigma_star_s[2] = Sigma_star_s[1] - quad_form_sym(Sigma_s[1],
phi_splusone_star[1]');
40   S_s[2] = sqrtm(Sigma_s[2]);
41   S_star_s[2] = sqrtm(Sigma_star_s[2]);
42   Gamma[2] = (phi_splusone[1] * Sigma_star_s[1])';
43
44   if (p == 1) {
45     phiGamma[1, p] = phi_splusone[1];
46     phiGamma[2, p] = Gamma[1];
47     return phiGamma;
48   } else {
49     array[p] matrix[m, m] store, store_star;
50     store[1] = phi_splusone[1];
51     store_star[1] = phi_splusone_star[1];
52     for (int splusone in 2 : p) {
53       array[splusone - 1] matrix[m, m] phi_s, phi_s_star;
54       matrix[m, m] Gamma_temp, matrix_sum;
55       phi_s = store[1 : (splusone - 1)];
56       phi_s_star = store_star[1 : (splusone - 1)];
57       phi_splusone[splusone] = mdivide_right(S_s[splusone] *
P[splusone],
58                                           S_star_s[splusone]);
59       phi_splusone_star[splusone] = mdivide_right(S_star_s[splusone] *
60                                           P[splusone]',
S_s[splusone]);
61
62       for (int i in 1 : (splusone - 1)) {
63         phi_splusone[i] = phi_s[i] - phi_splusone[splusone] *
phi_s_star[splusone - i];
64         phi_splusone_star[i] = phi_s_star[i] -
phi_splusone_star[splusone] * phi_s[splusone - i];
65       }
66       store[1 : splusone] = phi_splusone[1 : splusone];
67       store_star[1 : splusone] = phi_splusone_star[1 : splusone];
68       Sigma_s[splusone + 1] = Sigma_s[splusone] -
quad_form_sym(Sigma_star_s[splusone],
69
phi_splusone[splusone]');
70       S_s[splusone + 1] = sqrtm(Sigma_s[splusone + 1]);
71       Sigma_star_s[splusone + 1] = Sigma_star_s[splusone] -
72       quad_form_sym(Sigma_s[splusone],
73
phi_splusone_star[splusone]');

```

```
74      S_star_s[splusone + 1] = sqrtm(Sigma_star_s[splusone + 1]);
75      matrix_sum = phi_splusone[splusone] * Sigma_star_s[splusone];
76      for (int i in 1 : (splusone - 1)) {
77          Gamma_temp = Gamma[splusone - i + 1];
78          matrix_sum += phi_s[i] * Gamma_temp';
79      }
80      Gamma[splusone + 1] = matrix_sum';
81  }
82  phiGamma[1, : ] = phi_splusone;
83  phiGamma[2, : ] = Gamma[1 : p];
84  return phiGamma;
85  }
86  }
87  }
```

B Supplementary Material

The following table presents the coefficients of 5 stationary VARs simulated with 12 lags ($p = 12$) and one dimension ($m = 1$) using the procedure described in Section 4.3.2, and, for each experiment, it also displays the maximum absolute value of all eigenvalues. Those coefficients then simulate 5 datasets of 500 observations ($N = 500$) according to the procedure outlined in the same section. Note that since the maximum absolute eigenvalue is smaller than 1 for all experiments, all processes are indeed stationary:

Table 9: Coefficients of simulated stationary VAR processes with 12 lags ($p = 12$) and a single dimension ($m = 1$) across 5 experiments.

$m = 1$					
Lag	Exp. 1	Exp. 2	Exp. 3	Exp. 4	Exp. 5
1	1.68	1.04	-0.17	0.05	-0.87
2	-0.26	-0.23	-0.19	-1.09	-0.71
3	2.03	0.15	1.34	-0.28	0.34
4	-3.16	0.17	-0.17	-0.22	0.69
5	-2.10	-1.37	-0.16	-0.56	1.11
6	-0.22	2.15	0.07	0.11	0.99
7	2.33	-0.90	0.02	-0.78	0.26
8	-3.15	-0.01	0.10	0.13	-0.29
9	1.64	-0.04	-1.27	-0.76	-0.73
10	0.73	-0.56	0.32	-0.64	-0.07
11	-1.54	1.15	0.27	-0.25	-0.10
12	0.71	-0.62	0.83	-0.81	0.38
Max Eigenvalue Modulus	0.99	0.99	0.99	0.99	0.99

Figure 14 shows the trace plot for the thirteenth transformed partial autocorrelation coefficient for the posterior constructed using the extended MGP:

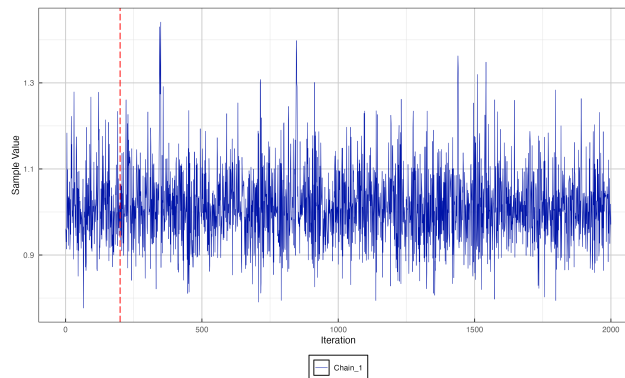


Figure 14: Trace plot for the single chain used to sample from the posterior distribution of the effective lag order; Extended MGP.

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